

Probing PAC1 receptor activation across species with an engineered sensor

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Reto B. Cola, Salome N. Niethammer, Preethi Rajamannar, Andrea Gresch, Musadiq A. Bhat, Kevin Assoumou, Elyse T. Williams, Patrick Hauck, Nina Hartrampf, Dietmar Benke, Miriam Stoeber, Gil Levkowitz, Sarah Melzer, Tommaso Patriarchi 

Institute of Pharmacology and Toxicology, University of Zürich, Zürich, Switzerland; • Medical University of Vienna, Center for Brain Research, Department for Neuronal Cell Biology, Vienna, Austria • Department of Molecular Neuroscience & Department of Molecular Cell Biology Weizmann Institute of Science, Israel • Department of Cell Physiology and Metabolism, University of Geneva, Geneva, Switzerland; • Department of Chemistry, University of Zürich, Zürich, Switzerland; • Neuroscience Center Zurich, University and ETH Zürich, Zürich, Switzerland;

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Abstract

Class-B1 G protein-coupled receptors (GPCRs) are an important family of clinically relevant drug targets that remain difficult to investigate via high-throughput screening and in animal models. Here, we engineered PAClight1_{P78A}, a novel genetically-encoded sensor based on a class-B1 GPCR (the human PAC1 receptor, hmPAC1R) endowed with high dynamic range ($\Delta F/F_0 = 1100\%$), excellent ligand selectivity and rapid activation kinetics ($\tau_{ON} = 1.15$ sec). To showcase the utility of this tool for in vitro applications, we thoroughly characterized and compared its expression, brightness and performance between PAClight1_{P78A} transfected and stably-expressing cells. Demonstrating its use in animal models, we show robust expression and fluorescence responses upon exogenous ligand application ex vivo and in vivo in mice, as well as in living zebrafish larvae. Thus, the new GPCR-based sensor can be used for a wide range of applications across the life sciences empowering both basic research and drug development efforts.

eLife assessment

This **fundamental** paper reports a new biosensor to study G protein-coupled receptor activation by the pituitary adenylyl cyclase-activating polypeptide (PACAP) in cell culture, ex vivo (mouse brain slices), and in vivo (zebrafish). **Convincing** data are presented that show the new sensor works, albeit at very high (non-physiological) concentrations of exogenous PACAP. The sensor has not yet been used to detect endogenously released PACAP, raising questions about whether the sensor can be used for its intended purpose. While further work must be pursued to achieve broad in vivo applications under physiological conditions, the new tool will be of interest to cell biologists, especially those studying the large GPCR family.

Introduction

Class-B1 G protein-coupled receptors (GPCRs) represent an important sub-group of peptide-sensing GPCRs, that are the focus of intense and rapidly-expanding drug development efforts^{1,2}, driven by extremely successful examples of peptide agonists used in the clinical treatment of metabolic human diseases, such as type-2 diabetes and obesity³. One such peptide-GPCR system that has shown growing potential for targetability in the treatment of human disorders is the Pituitary Adenylate Cyclase Activating Peptide (ADCYAP1 or PACAP) and its receptors. PACAP is an endogenous 38-amino acid peptide that is amongst the most phylogenetically-conserved peptides⁴. Its shorter C-terminally-truncated form (i.e. PACAP₁₋₂₇) has 68% homology with the vasoactive-intestinal-peptide (VIP)⁵. In fact, VIP and PACAP share a subfamily of class-B1 G protein-coupled receptors (GPCRs), of which the two receptors VPAC1 and VPAC2 can both be equipotently activated by VIP and PACAP⁵. The third receptor in this subfamily, i.e. the PAC1 receptor (PAC1R, also known as ADCYAP1R1), however, has a reported affinity for PACAP that is 100-1000-fold higher than its affinity to VIP⁵. The tissue distribution of PACAP and its receptors is widespread and they can be found throughout the central and peripheral nervous system, the immune system, in endocrine glands as well as in other organ systems and in many cancerous tissues⁵⁻⁹. A large body of evidence has linked the PACAP/PAC1R system to protective functions in the nervous and immune systems, as well as to stress- and anxiety-related behaviors (in particular post-traumatic stress disorder (PTSD)), migraine, nociception, thermoregulation, sleep/wake cycles, and reproductive functions^{5,8,10}, making it a peptide signaling system of high clinical relevance.

Recently, genetically-encoded GPCR-based sensors have been developed that enable the direct optical detection of GPCR activation by agonist ligands with high sensitivity and spatiotemporal resolution¹¹⁻¹⁹. While these tools hold great potential for drug development, pharmacology and neuroscience applications, only a few of them are built from class-B1 peptide-sensing GPCRs^{18,20}. Thus, the development of new biosensors based on peptide-sensing GPCRs would greatly benefit the community and serve as a powerful resource for drug screening and life sciences.

In this work we engineered PAClight1_{P78A}, an ultrasensitive indicator based on the human PAC1 receptor (hmPAC1R). To establish this as a tool for peptide drug screening, we thoroughly characterized its dynamic range, as well as its optical, kinetic, signaling and pharmacological properties *in vitro*. We further generated stable cell lines and compared them to transfected cells for ligand-induced fluorescence responses using flow cytometry. Additionally, we tested the potential of PAClight1_{P78A} as a tool to study ligand binding and diffusion in animal models. To this end, we examined the sensitivity and specificity of PAClight1_{P78A} in acute mouse brain slices. Moreover, we verified ligand detection *in vivo* using fiber photometry recording and intracerebral microinfusions in behaving mice.

Finally, we demonstrate that the sensor expresses well and produces a large fluorescent response to PACAP₁₋₃₈ following microinjection in the brain of living zebrafish larvae, opening new opportunities for the development and testing of drugs targeting this receptor in the central nervous system.

Overall, our new sensor expands the class of genetically-encoded optical tools that can be used for GPCR-targeted HTS assays, reducing the demands on time and costly reagents, and provides new opportunities for functionally testing drugs that target the PAC1R pathway directly in living animals.

Results

Development of an ultrasensitive PAC1R-based sensor

To develop a highly-sensitive indicator of hmPAC1R activation, we followed a protein engineering approach that we and others recently established^{11,14,17}. We used the human PAC1Rnull splice isoform as a protein scaffold (hmPAC1R) and constructed an initial sensor prototype in which we replaced the entire third intracellular loop (ICL3, residues Q336-G342) with a module containing circularly-permuted green fluorescent protein (cpGFP) from dLight1.3b¹¹ (**Extended Data Fig. 1a**). This initial prototype construct, named PAClight0.1, was expressed on the surface of HEK293T cells, but showed clear intracellular retention and a very weak average fluorescent response to bath application of PACAP₁₋₃₈ ($\Delta F/F_0 = 43.4\%$, **Extended Data Fig. 1b**). Using this sensor as template, we conducted a small-scale screen to identify the optimal insertion site by reintroducing amino acids from the original ICL3 of hmPAC1R on both sides of the cpGFP-module. This led us to the identification of a second mutant, in which Q336 was reintroduced before the cpGFP-module, that showed improved average fluorescent response to the ligand ($\Delta F/F_0 = 343\%$, **Extended Data Fig. 1c**). To improve membrane expression of the PAClight mutants, we next investigated the effect of point mutations on their C-terminus, via site-directed mutagenesis. Mutation into alanine of residue T468, a reported post-translational modification (PTM) site (www.phosphosite.org) on the C-terminus of the receptor²¹, improved the overall surface expression of the sensor in HEK293T cells and further increased the observed average fluorescent response ($\Delta F/F_0 = 516\%$, **Extended Data Fig. 1f**). To further improve the fluorescent response of PAClight, we screened a subsequent library of sensor variants with mutations targeted to the second intracellular loop (ICL2). A similar approach was previously shown to boost the fluorescent response of other GPCR-based sensors, for example dLight1.3b or OxLight1^{11,17}. Our functional screen of a library of ICL2 mutants led to the identification of one variant comprising three point mutations (F259K/F260K/P261K) that displayed a largely improved fluorescent response to PACAP₁₋₃₈ ($\Delta F/F_0 = 942\%$, **Extended Data Fig. 1d**). Upon introduction of the C-term T468A mutation on the ICL2 F259K/F260K/P261K background, we obtained an improved version of the sensor, named PAClight1, with excellent surface expression and fluorescent response ($\Delta F/F_0 = 1037\%$, **Extended Data Fig. 1g**). Given the high degree of sequence and structural similarity, as well as the broad activation of endogenous VPAC receptors by vasoactive intestinal peptide (VIP) and PACAP²², we next asked whether the hmPAC1R-based sensor would also respond to VIP. Indeed, application of a high concentration of VIP onto sensor-expressing cells caused a large fluorescent response, corresponding to more than half of the response to PACAP₁₋₃₈ ($\Delta F/F_0 = 654\%$, **Extended Data Fig. 2c and 2d**). In an effort to eliminate the response to VIP and obtain a PACAP-specific sensor, we screened a small library of sensors containing single- point mutations into alanine that were inspired by simulations on the binding free energy of PACAP and VIP on the PAC1R²³. Furthermore, at this stage we included the naturally occurring splice mutant PAC1s into the extracellular ligand-binding domain (ECD) of the sensor (this is a short ECD mutant reported to have lower affinity towards VIP⁹), as well as two structure-guided point-mutants. Through this screening, we identified a single mutation (P78A) that abolished the sensor response to VIP while leaving unaltered the response to PACAP₁₋₃₈ (**Extended Data Fig. 2c and 2d**). The final sensor construct, named PAClight1_{P78A} (**Figure 1a,b**), includes all the above-mentioned mutations and combines a very large dynamic range with excellent PACAP selectivity and good membrane expression, and was thus selected for further *in vitro* characterization.

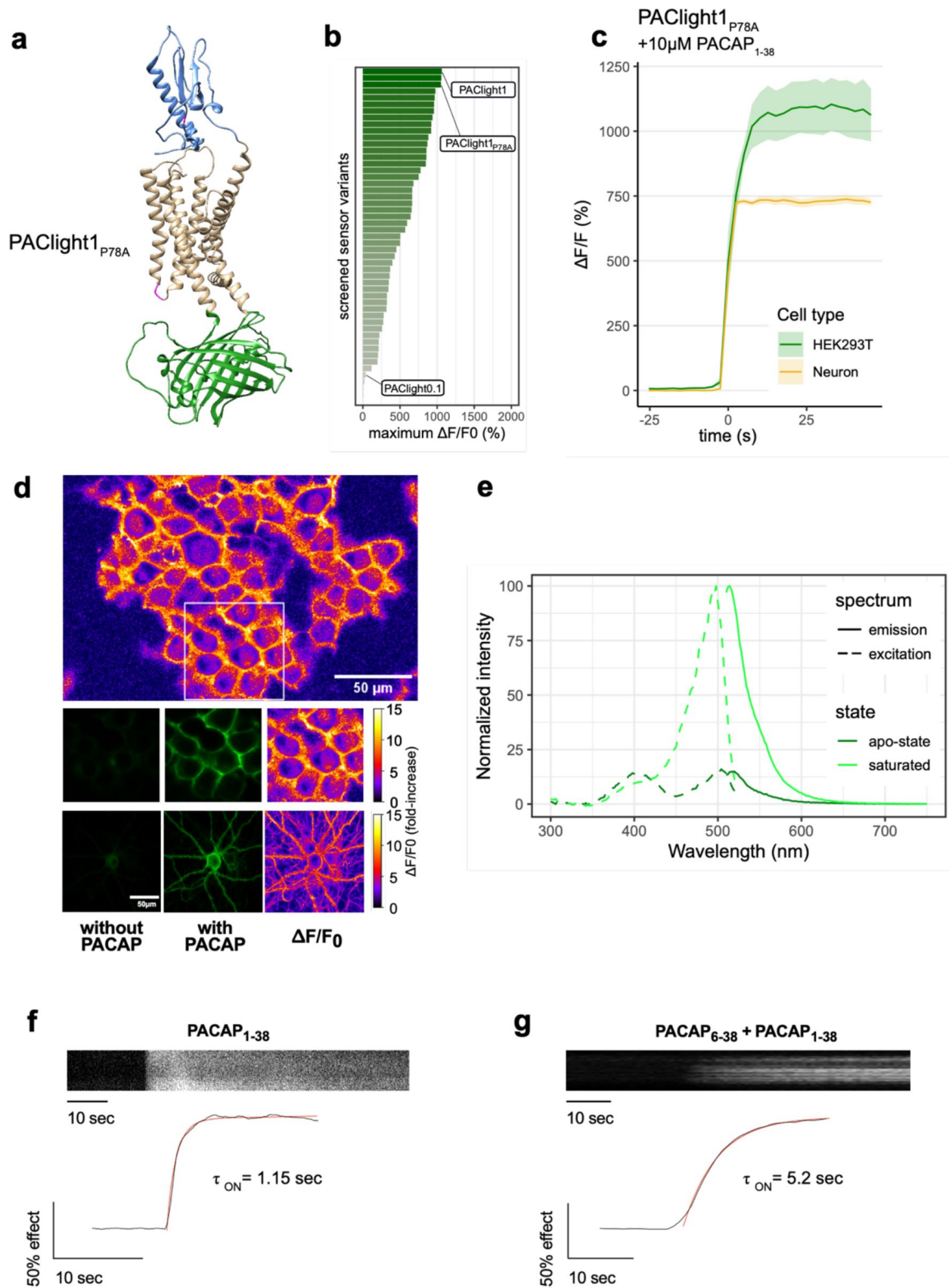


Figure 1

Design and in vitro optical properties of PAClight1_{P78A}

a) Protein structure of PAClight1_{P78A} as predicted by AlphaFold2⁵⁰. The transmembrane and intracellular domain of the PAC1Rnull backbone is depicted in beige. The extracellular domain, which is crucial for ligand specificity and affinity, is colored in blue. The cpGFP module replacing the 3rd intracellular loop is colored in green. Point-mutations inserted in the 2nd intracellular loop as well as on the extracellular domain are depicted in magenta. **b)** Bar chart depicting the maximum dynamic range obtained for all tested mutants with acceptable expression on the plasma membrane. Bars are ordered by dynamic range and color coded by the average of the maximum dynamic range recorded. The bars representing the prototype variant PAClight_V0.1, PAClight1_{P78A} and PAClight1 are indicated. **c)** Representative line plots depicting the maximum activation (mean $\pm$ SEM) of PAClight1_{P78A} expressed in HEK293T cells and rat primary neurons. Upon bath application of 10 μ M PACAP₁₋₃₈, PAClight1_{P78A} reaches a $\Delta F/F_0$ of 1055% in HEK293T cells (n=1, 5 ROIs) and a $\Delta F/F_0$ of 729% in rat primary neurons (n=1, 8 ROIs). Confocal image acquisition was performed at a frame rate of 1 frame/2.53s ($\sim$ 0.4Hz). **d)** Representative examples of the expression of PAClight1_{P78A} in HEK293T cells (low magnification overview and middle row of images) and rat primary neurons (bottom row) with pixel-wise quantification and depiction of the dynamic range in $\Delta F/F_0$ upon bath application of 10 μ M PACAP₁₋₃₈. Top: Overview micrograph of HEK293T cells color coded in pixel-wise $\Delta F/F_0$. The white square represents the ROI represented below in the middle row of images. Middle: Selected ROI before (left) and after (middle and right) peptide application. Bottom: Rat primary neuron expressing PAClight1_{P78A} before (left) and after (middle and right) peptide application. The color bars represent the look-up table used for visualization of the pixel-wise $\Delta F/F_0$. **e)** One-photon excitation (dashed lines) and emission (solid lines) spectra of PAClight1_{P78A} in the absence (apo-state, dark green) and the presence (saturated state, light green) of 10 μ M PACAP₁₋₃₈. The excitation maximum of the saturated state is at 498nm. The isosbestic point is at 420nm. Emission maximum in of the saturated state is at 514nm. **f)** Activation kinetics of PAClight1_{P78A} upon application of PACAP₁₋₃₈ (10 μ M) measured via timelapse imaging. A representative kymograph of sensor fluorescence on the surface of a HEK293 cell is shown on top. The normalized fluorescence response trace is shown at bottom along with the calculated one-phase association curve fit and activation time constant. The trace shown is the average of 3 independent experiments. **g)** Same as **f)** but in the presence of PACAP₆₋₃₈ (10 μ M) in the bath.

In vitro characterization of PAClight1_{P78A}

The newly developed PAClight1_{P78A} sensor displays an average fluorescent response of 1109% $\Delta F/F_0$ in HEK293T cells (n=3, 5 ROIs each) to bath application of 10 μ M PACAP₁₋₃₈ (**Figure 1b-d**). We verified that sensor expression and function is not drastically affected by the cell type in which it is expressed by testing it in primary cultured neurons. Neurons were virally transduced using an adeno-associated virus (AAV) for expressing the PAClight1_{P78A} sensor under control of a human synapsin-1 promoter. 2-3 weeks after transduction we verified excellent expression of the probe on the plasma membrane of the neurons with no appreciable intracellular retention. Under these conditions, the sensor showed a fluorescence response of approximately 730% $\Delta F/F_0$ upon bath application of the PACAP₁₋₃₈ ligand (**Figure 1c,d**).

We next set out to determine the sensor's excitation and emission spectra in vitro in HEK29T cells (**Figure 1e**). The excitation maxima in the absence and presence of 10 μ M PACAP₁₋₃₈ were identified at 504nm and 498nm respectively. The isosbestic point at which the excitation is independent of the absence or presence of PACAP₁₋₃₈ is located at 420nm. The maxima for the emission spectra in the absence of PACAP₁₋₃₈ was identified at 520nm and in the presence of PACAP₁₋₃₈ at 514nm.

Our previous work on the development of another class-B1 sensor based on the GLP1 receptor, led us to discover that the kinetics of the sensor's response can be used to infer the occupancy of the ECD by an antagonist peptide²⁴. Given that all class-B1 GPCRs share a similar ECD high-affinity ligand binding mechanism, we performed similar experiments to determine whether PAClight1_{P78A} could also be used in a similar manner. To do so, we monitored the sensor's response during application of PACAP₁₋₃₈ alone or in the presence of PACAP₆₋₃₈, an antagonist peptide, in the buffer surrounding the cells. We then determined the activation time constant of the fluorescent response in both conditions. The PAClight1_{P78A} response was strikingly slower (approximately 4-fold) in the presence of the extracellular antagonist, and was in the range of 1 second in its absence (**Figure 1f,g**). Thus, the kinetics of PAClight1_{P78A} response could be used to investigate or screen for factors that influence the ECD-PACAP interaction and potentially the speed of signal transduction through conformational activation of the receptor.

To investigate the stability of the fluorescent response to bath application of PACAP₁₋₃₈, a long-term imaging experiment was performed at room temperature with 1 frame (1024x1024 pixels, 2x line-averaging) acquired every minute over the time course of 150 min (**Figure 2a**). After bath application of 200nM PACAP₁₋₃₈ (slow diffusion of the peptide with 10x dilution from 2μM to 200nM) there was no decrease in signal, nor any internalization observed. After >100 frames of acquisition, we applied a saturating bolus of the peptide PAC1R-antagonist Max.d.4. Within 45-50 frames after application of the PAC1R antagonist, the signal intensity steadily decreased and started to plateau slightly above baseline levels. This indicates, that Max.d.4 can outcompete PACAP₁₋₃₈ at PAClight1_{P78A}, albeit not to a full extent under the specified experimental conditions.

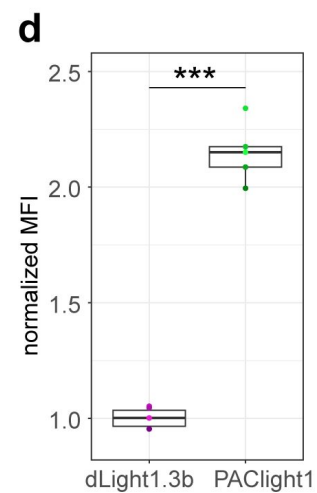
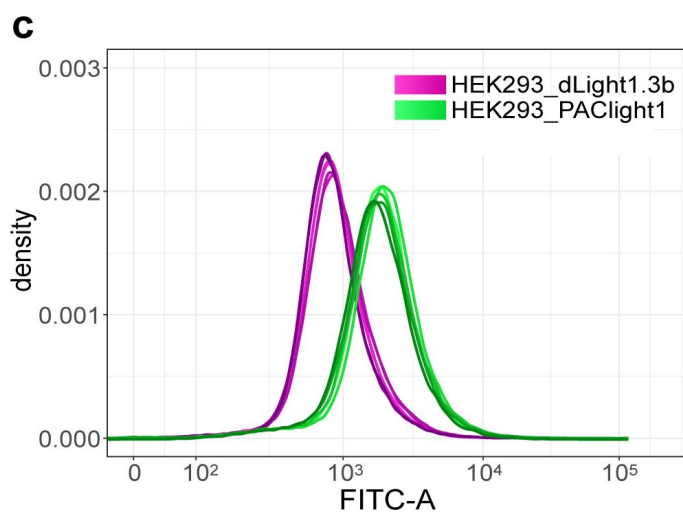
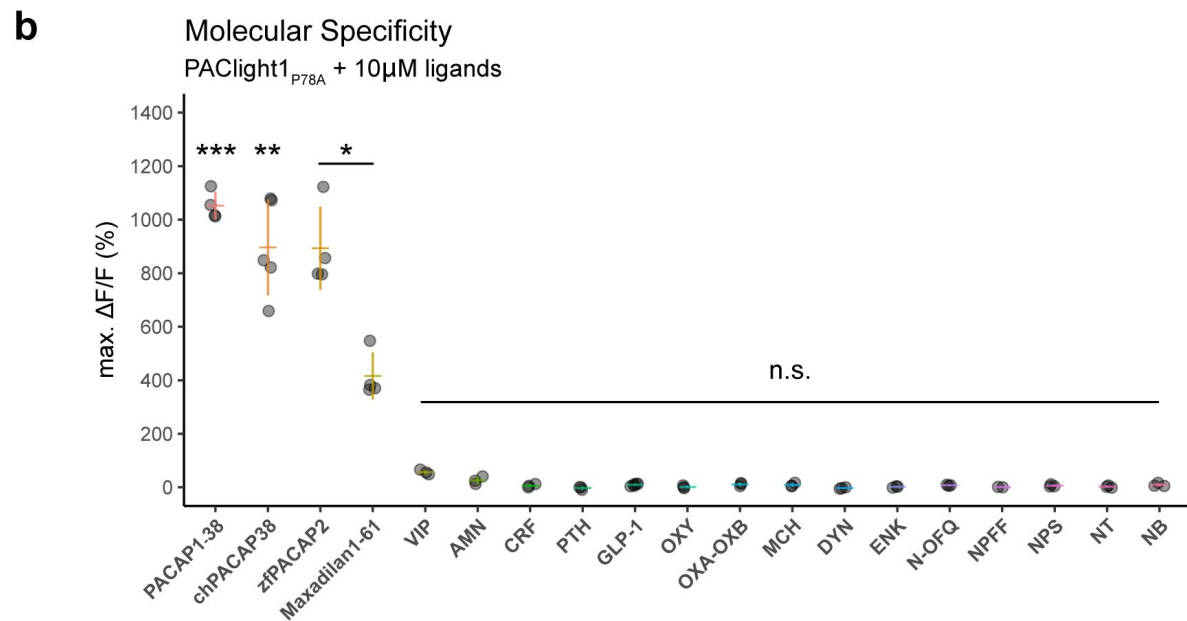
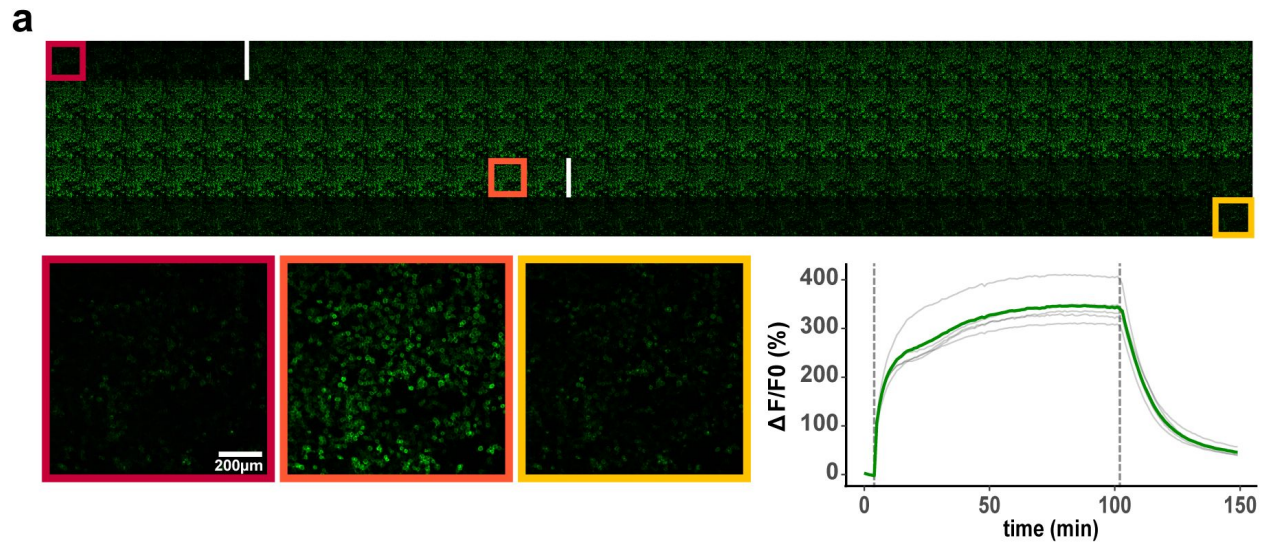


Figure 2

Pharmacological characterization of PAClight1_{p78A}

a) PAClight1_{p78A} fluorescent response to 200 nM PACAP₁₋₃₈ bath application over extended periods at room temperature. No internalization of the sensor expressed on the plasma membrane is observed throughout the full time course of >90 minutes. The fluorescent response of PAClight1_{p78A} to PACAP₁₋₃₈ can be reversed with competitive binding of the peptidergic PAC1R antagonist Max.d.4 (a Maxadilan derivative). Please note that the seemingly slow activation rate of PAClight1_{p78A} in this experiment is due to the experimental setup (see methods section) and slow diffusion of PACAP₁₋₃₈ throughout the well. Top: Downscaled gallery view of the acquired time series (total of 150 frames; 1 frame/minute) from top-left to bottom-right. Application time points of PACAP₁₋₃₈ (after frame Nr. 5) and Max.d.4 (after frame Nr. 103) are depicted as white vertical lines between the frames. The red, orange, and yellow rectangles indicate the frames used for representative higher magnification inserts shown below (bottom left). Bottom right: Line plot depicting the time course of the PAClight1_{p78A} response across 5 rectangular ROIs distributed across the whole field of view. **b)** PAClight1_{p78A} is highly specific to PACAP and does not respond to VIP. PAClight1_{p78A} can be activated by PACAP homologues found in the chicken (gallus gallus, chPACAP38) as well as in the zebrafish (danio rerio, zfPACAP2). PAClight1_{p78A} is further partially activated by the PAC1R specific ligand Maxadilan₁₋₆₁, which is expressed endogenously in the sand fly (*Lutzomyia longipalpis*) salivary gland. None of the other tested class A and B1 GPCR ligands was found to activate PAClight1_{p78A}. Abbreviations: AMN amnesiac, CRF corticotropin- releasing factor, PTH parathyroid hormone, GLP-1 glucagon-like peptide, OXY oxytocin, OXA- OXB orexin-A and -B, MCH melanin-concentrating hormone, DYN dynorphin, ENK enkephalin, N-OFQ nociception, NPFF neuropeptide FF, NPS neuropeptide S, NT neurotensin, NB neuromedin B. Single data points represent one replicate average obtained from 5 ROIs per replicate. The extent of the colored vertical bar represents 1 standard deviation. The y-axis location of the colored horizontal bar indicates the average across all replicates. Number of replicates per ligand: n=5 for chPACAP38 and GLP-1; n=4 for PACAP₁₋₃₈, zfPACAP2, and Maxadilan₁₋₆₁; n=3 for VIP, AMN, CRF, PTH, OXY, OXA-OXB, MCH, DYN, ENK, N-OFQ, NPS, NT, and NB; n=2 for NPFF. Asterisks represent statistical significance of Hochberg-corrected *p* values of multiple one-sample *t* tests. **c)** PAClight1 has significantly brighter baseline fluorescence than the dopamine sensor dLight1.3b. The peak of the dLight1.3b FITC-A density curve coincides with the start of the uphill slope of the PAClight1 FITC-A density curve. Note the bi-exponential scaling of the x-axis. n=6 for dLight1.3b, n=5 for PAClight1. **d)** Quantification of the average of the median fluorescence intensity (MFI) across all replicates and normalization to the group MFI of dLight1.3b show a 2.15-fold increased basal brightness of PAClight1 over dLight1.3b ($t_{(4,74)} = -19.254$, $p < 0.0001$, 95% CI [-1.30, -0.99], two-sided two-sample Welch's *t* test).

Next, we screened a range of different peptides at saturating concentrations (10 μM) for potential activation of the PAClight1_{p78A} sensor (**Figure 2b**). Mammalian PACAP₁₋₃₈ displayed the strongest potency in PAClight1_{p78A} activation (1052% ΔF/F₀, $t_{(3)} = 40.31$, 95% CI [9.69, 11.35], $p < 0.001$). Because the amino acid sequence of PACAP is rather well conserved throughout phylogeny, we also tested chicken PACAP₁₋₃₈ (chPACAP38), as well as the zebrafish PACAP2₁₋₂₇ (zfPACAP2) for potency on the PAClight1_{p78A} sensor. Both of these homologues of the mammalian PACAP₁₋₃₈ activated the PAClight1_{p78A} sensor with strong but slightly reduced potency (chPACAP38: 896% ΔF/F₀, $t_{(4)} = 11.17$, 95% CI [6.73, 11.19], $p = 0.0066$; zfPACAP2: 898% ΔF/F₀, $t_{(3)} = 11.49$, 95% CI [6.46, 11.41], $p = 0.024$). Furthermore, we also screened the sand fly salivary gland derived peptide Maxadilan, which was previously identified to be a specific ligand of the PAC1R but not to the VPAC1 and VPAC2 receptors²⁵. Consistent with the reported activity of Maxadilan on the hmPAC1Rnull receptor, we detected activation of PAClight1_{p78A} by Maxadilan, however, with lower potency than mammalian PACAP₁₋₃₈ (416% ΔF/F₀, $t_{(3)} = 9.47$, 95% CI [2.76, 5.56], $p = 0.04$). The reduced potency of Maxadilan for PAClight1_{p78A} might be a consequence of the ligand specificity determining point-mutation (P78A) introduced into the extracellular domain of the PAClight1_{p78A} sensor. The drosophila gene *amn* (amnesiac) was previously shown to be homologous to the mammalian gene encoding PACAP (*Adcyap1*) and *amn* drosophila mutants show memory impairments similar to rodent PACAP/PAC1R mutants²⁶. We therefore also tested

whether this insect peptide encoded by *amn* would activate PAClight1_{P78A}, but no significant activation above baseline was observed (26% $\Delta F/F_0$, $t_{(2)} = 3.2$, 95% CI [-0.09, 0.61], $p = 0.69$). As shown above (**Extended Data Fig. 2c** and **2d**) PAClight1_{P78A} was specifically optimized for VIP non-responsive properties. Therefore, the response of PAClight1_{P78A} to VIP in this specificity screen was also not significantly above baseline (57% $\Delta F/F_0$, $t_{(2)} = 11.09$, 95% CI [0.35, 0.79], $p = 0.12$). Other peptides tested in this screen included other class B1 GPCR ligands (corticotropin-releasing factor (CRF), parathyroid hormone (PTH), and glucagon-like peptide 1 (GLP-1)), as well as some peptidergic class A GPCR ligands (oxytocin (OXY), orexin-A and -B (OXA-OXB), melanin-concentrating hormone (MCH), dynorphin (DYN), enkephalin (ENK), nociception (N-OFQ), neuropeptide FF (NPFF), neuropeptide S (NPS), neurotensin (NT), and neuromedin B (NB)). None of these neuropeptides activated the PAClight1_{P78A} sensor above baseline level (**Figure 2b**). Taken together, these results highlight the broad potential applicability of the PAClight1_{P78A} sensor for use in model systems across the phylogenetic tree, as well as its high selectivity for PACAP ligands over VIP and other peptide GPCR ligands.

In order not to induce artificial PACAP signaling and potentially interfere with downstream readouts when using PAClight1_{P78A}, it is important to verify that the sensor does not recruit G-proteins and/or β -arrestin. To monitor the capacity of PAClight1_{P78A} to engage these intracellular signaling partners, we performed split NanoLuc complementation assays as in our previous work^{17,19}, using either PAClight1_{P78A}-SmBiT or, as a positive control, PAC1R-SmBiT fusion constructs, together with LgBiT-miniGs (**Extended Data Fig. 3a**), LgBiT-miniGsq (**Extended Data Fig. 3b**), or LgBiT- β -arrestin2 (**Extended Data Fig. 3c**). As expected, we observed significant miniGs, miniGsq, and β -arrestin2 recruitment to the wild-type PAC1R upon activation with 1 μ M of PACAP₁₋₃₈. Yet, we did not detect recruitment of either miniGs, miniGsq, or β -arrestin2 in PAClight1_{P78A}-expressing cells upon stimulation with PACAP₁₋₃₈. These results indicate that expression of PAClight1_{P78A} does not artificially induce PACAP-mediated intracellular signaling and is not likely to interfere with endogenous signaling pathways.

Development of non-responsive PAClight1 control sensors

When employing GPCR-sensors in intact living tissue (e.g., when used in animal models) it is often desirable to make use of an appropriate control sensor, in which ligand-binding is abolished by virtue of one or more point mutations in the GPCR binding pocket. To engineer such control sensors for our PAClight1_{P78A} and PAClight1 sensors, we targeted key residues in the hmPAC1R that interact with residue D3 in PACAP. The carboxylic group of D3 was shown to be crucial for binding affinity and biological activity of PACAP to all three receptor types (i.e. PAC1, VPAC1, and VPAC2)^{27,28}. Furthermore, the cryo-EM structure of the hmPAC1R in interaction with PACAP identified residues Y161 and R199 of the hmPAC1R to interact with residue D3 of PACAP²². Residue Y161 forms a hydrogen bond with D3, while R199 forms an electrostatic interaction with D3 (**Extended Data Fig. 4a** and **4b**). To abolish these interactions, we first mutated R199 into alanine (R199A) on the backbone of PAClight1. This single point mutant showed drastically reduced average fluorescent response (PAClight1: 1075% $\Delta F/F_0$, PAClight1_R199A: 41.9% $\Delta F/F_0$, **Extended Data Fig. 4c**) to 10 μ M PACAP₁₋₃₈. To further abolish the remaining response, we additionally mutated Y161 into alanine (Y161A) on the previous backbone (i.e. PAClight1_R199A_Y161A) and named the construct “PAClight1-ctrl”. This completely reduced the fluorescent response to PACAP₁₋₃₈ to baseline levels (PAClight1-ctrl: 2.87% $\Delta F/F_0$). Next, we cloned these two point mutations into the backbone of PAClight1_{P78A} and observed an equally abolished fluorescent response to PACAP₁₋₃₈ (PAClight1_{P78A}: 1034% $\Delta F/F_0$, PAClight1_{P78A}-ctrl: 10.44% $\Delta F/F_0$, $t_{(4.03)} = 37.89$, $p < 0.0001$, 95% CI [1098.63, 949.02], two-sided two-sample Welch's t test, **Extended Data Fig. 4d**) in transfected HEK293T cells, as well as in virally-transduced rat primary neuron cultures (**Extended Data Fig. 4f**). No drastic differences in basal brightness were observed between PAClight1_{P78A} and PAClight1_{P78A}-ctrl constructs (**Extended Data Fig. 4e**). In summary, we have developed double point-mutant sensor constructs for both PAClight1_{P78A} and PAClight1 with fully abolished responses to bath application of saturating concentrations of PACAP₁₋₃₈.

Comparison between transient and stable expression of the sensor

During the development and validation of PAClight1_{P78A}, we noticed higher basal brightness levels of PAClight1_{P78A} compared to other GPCR-based fluorescent sensors. To obtain a quantitative comparison, we decided to produce a stable T-Rex HEK293 cell line for inducible expression of PAClight1 (mutant missing P78A ECD mutation). We then used flow cytometry to record multiple replicates of 100,000 cells, and compared their fluorescence intensity to that of a similarly generated cell line expressing the indicator dLight1.3b, which we previously described²⁹ (Figure 2c-d). Visualization of the fluorescent readouts clearly showed a shift of the density curves towards higher fluorescence for HEK293_PAClight1 cells compared to the HEK293_dLight1.3b. Statistical comparison of the distributions of the two cell lines shows 2.15- fold higher basal brightness of PAClight1 over dLight1.3b ($t_{(4,74)} = -19.254$, $p < 0.0001$, 95% CI [- 1.30, -0.99], two-sided two-sample Welch's t test, Figure 2d).

Among the most important potential advantages of using stable cell lines expressing GPCR sensors for pharmacological assays are the homogeneity of expression, reproducibility, and ease of use. To verify that this is indeed the case, we analyzed PAClight1_{P78A} expressing stable cells and compared them to transfected cells using flow cytometry. While a large proportion of cells from the transfected condition expressed the sensor at very low levels or not at all (i.e., were left shifted in the density plot), we also observed a significant proportion of very bright and strongly expressing cells from within the same condition. In fact, there is higher abundance of very bright and strongly expressing cells in the transfected condition than in the induced stable cell line condition (Figure 3a). The combination of the higher abundance of very low expression levels and very high expression levels within the transfected condition leads to a significantly increased variability of observed expression levels compared to the induced stable cell line condition. The median of the standard deviation in the FITC-A channel across the titration series was 62.6% smaller in the stable PAClight1_{P78A} relative to the transfected PAClight1_{P78A} condition ($p = 0.0142$, Figure 3b). The variability of cells transfected with PAClight1 was very comparable and was only 8.6% lower than in the PAClight1_{P78A} transfected condition ($p = 0.5966$, Figure 3b). The difference of 8.6% (even though statistically insignificant) might partially be explained by slightly different performance of PAClight1_{P78A} vs. PAClight1 in dynamic range. P values represent the result of Dunnett's test for correction of multiple comparison that was performed after the statistically significant results of an omnibus ANOVA ($F_{(2,30)} = 4.321$, $p = 0.0224$).

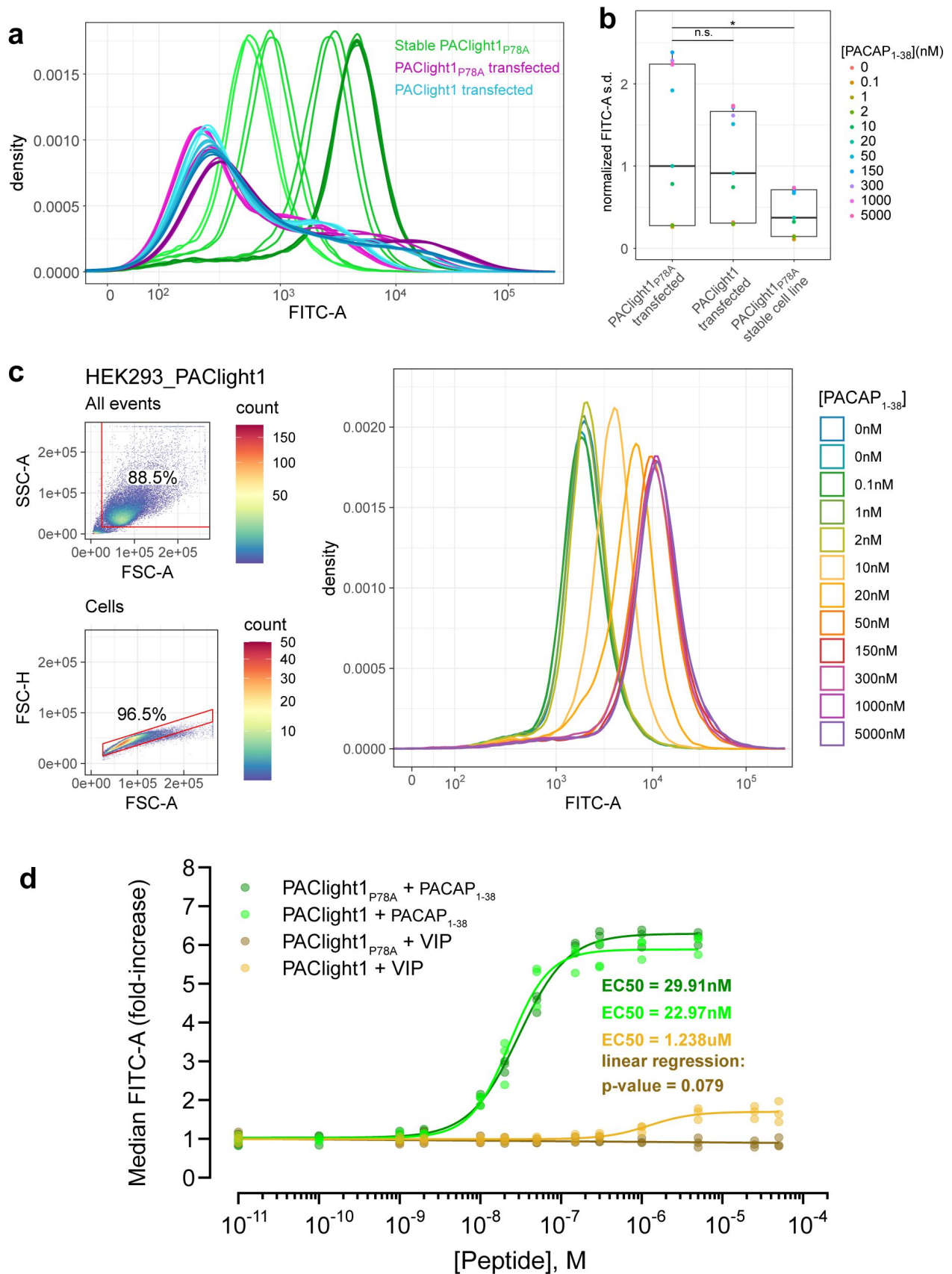


Figure 3

Comparison between transient and stable PAClight1_{p78A} expression

a) A direct flow cytometric comparison between transfected HEK293T cells and stable and inducible HEK293_PAClight1_{p78A} cells highlights the improved homogeneity of the distribution of expression levels in the newly generated stable cell line. Note the increased number of cells on both the low and high extremes in the transfected populations over the stable cell line population. Data from a full PACAP₁₋₃₈ titration series for each condition is presented. The overall variability in expression levels in the PAClight1_{p78A}⁻ and PAClight1-transfected condition is very similar. **b)** The standard deviation of each density curve (normalized to the median standard deviation of the transfected PAClight1_{p78A} condition) is plotted by condition and concentration. The median standard deviation of the stable HEK293_PAClight1_{p78A} condition is reduced to only 37.4% of the transfected PAClight1_{p78A} condition (Dunnett's test $p = 0.0142$). The median standard deviation varies little between transfected PAClight1_{p78A} and PAClight1 cells (8.6% lower in PAClight1, Dunnett's test $p = 0.5966$). **c)** A representative example of a PACAP₁₋₃₈ titration series on stable HEK293_PAClight1 cells. Top left: Gating strategy used to gate cells by the area of the side scatter (SSC-A) vs. the area of the forward scatter (FSC-A). Bottom left: Gating strategy used to gate singlets (within the cells gate) by the height of the forward scatter (FSC-H) vs. the FSC-A. Right: Density curves of the FITC-A channel obtained from the singlet gate. **d)** Dose-response curves obtained from VIP and PACAP₁₋₃₈ titrations on stable HEK293_PAClight1_{p78A} and stable HEK293_PAClight1 cells. PAClight1_{p78A} and PAClight1 both have high affinities for PACAP₁₋₃₈ (PAClight1_{p78A} EC₅₀ = 29.91nM, PAClight1 EC₅₀ = 22.97nM, $n = 3$). While PAClight1 still shows a response to higher concentrations of VIP (EC₅₀ = 1.24μM), PAClight1_{p78A}'s response to VIP is completely abolished up to concentrations of 50μM VIP ($F_{(1, 33)} = 3.28$, $p = 0.079$, adj. R² = 0.06). Data in **a-c** is derived from 100K original events recorded for each concentration and construct. Data in **d** is derived from 100K recorded events across $n = 3$ of each titration series.

With the stable cells and their improved homogeneity of expression levels, we then determined the affinities of PAClight1_{p78A} and PAClight1 towards PACAP₁₋₃₈ and VIP. For each peptide concentration of a titration series 100,000 events were acquired, gated for cell clusters and for singlets. The FITC-A median fluorescence intensity (MFI) of the events in the singlet gate were used for dose-response fitting. A representative PACAP₁₋₃₈ titration series replicate on stable HEK293_PAClight1 cells is shown in **Figure 3c**. The titration dataset reveals a slightly lower affinity of the PAClight1_{p78A} sensor (EC₅₀ = 29.91nM) than the PAClight1 sensor (EC₅₀ = 22.97nM) towards PACAP₁₋₃₈ (**Figure 3d**). The affinity of the PAClight1 sensor (non-specific mutant) towards VIP (EC₅₀ = 1.238μM) is much lower than its affinity towards PACAP₁₋₃₈ (**Figure 3d**). This 100-1000x difference is well in accordance with previous reports on the differential affinities of VIP and PACAP₁₋₃₈ on the PAC1R^{30,31}. We were not able to detect any response of PAClight1_{p78A} to the addition of VIP up to a concentration of 50μM (linear regression $p = 0.079$, **Figure 3d**), corroborating the specificity of the PAClight1_{p78A} sensor. Taken together, these data show that the PAClight1_{p78A} and PAClight1 sensors display very high affinity towards PACAP₁₋₃₈, with a slightly lower affinity of PAClight1 compared to PAClight1_{p78A} as the result of the point mutation on the extracellular domain of PAClight1_{p78A}.

Characterization of PAClight1_{p78A} and PAClight1_{p78A}-ctrl in model organism systems

Since PAClight1_{p78A} showed excellent expression and response properties in neuronal cell cultures, we next tested whether the sensor can be established as a tool to test ligand binding in intact neuronal circuits of mammalian model systems. Publicly available RNAseq databases (*in situ* hybridization atlas from the Allen Institute: alleninstitute.org) and previous work³² show strong expression of PACAP (*Adcyap1*) and PACAP receptors (*Adcyap1r1*) in the cerebral cortex and hippocampus of mice and humans, suggesting these structures as interesting candidates for drug

targeting. We tested the expression, sensitivity, and specificity of PAClight1_{P78A} in acute mouse brain slices that provide a physiological environment to examine the actions of pharmacological agents within specific brain areas³³.

AAVs encoding the PAClight1_{P78A} or PAClight1_{P78A}-ctrl sensor under control of a human synapsin-1 promoter were stereotactically injected into the neocortex and hippocampus of adult mice (**Figure 4a**). 4 weeks later, virus expression in the injection site was validated by histology (**Figure 4b**). PAClight1_{P78A} and PAClight1_{P78A}-ctrl were efficiently expressed in neuronal cell bodies, axons and dendrites (**Figure 4c**, **Extended Data Figure 5**).

To investigate the sensor response dynamics and its sensitivity, acute brain slices expressing PAClight1_{P78A} or PAClight1_{P78A}-ctrl were prepared and PACAP₁₋₃₈ was bath-applied at concentrations ranging from 0nM to 3000nM. We observed a dose-dependent fluorescence increase to PACAP₁₋₃₈ using PAClight1_{P78A}, but not PAClight1_{P78A}-ctrl (**Figure 4d**). PAClight1_{P78A} fluorescence increased by +6.8 and +17.5% $\Delta F/F_0$ when bath-applying 300 and 3000 nM of PACAP₁₋₃₈ respectively (**Figure 4d**). Comparing PAClight1_{P78A} fluorescence with PAClight1_{P78A}-ctrl fluorescence revealed a statistically significant difference when bath-applying 3000 nM of PACAP₁₋₃₈ (**Figure 4d**).

To assess the specificity of PAClight1_{P78A} to PACAP₁₋₃₈ in comparison to other neuropeptides in acute brain slices, we recorded its response to 3 μ M VIP, corticotropin-releasing factor (CRF), and enkephalin (ENK). PAClight1_{P78A} clearly increased its fluorescence in response to 3 μ M PACAP₁₋₃₈ (+17.5% peak $\Delta F/F_0$), but not to VIP (-5.5% peak $\Delta F/F_0$), CRF (-4.3% peak $\Delta F/F_0$), or ENK (+1.4% peak $\Delta F/F_0$) (**Figure 4e**).

In conclusion, our data show that in acute mouse brain slices, PAClight1_{P78A} detects concentrations of >300nM of PACAP₁₋₃₈ in superfused bath application, while not reacting to other neuropeptides tested.

In vivo PACAP detection in behaving mice

To test whether PAClight1_{P78A} can be used to characterize ligand binding and diffusion in vivo in behaving mice, we implanted fiberoptic cannula into the neocortex of mice expressing PAClight1_{P78A} or PAClight1_{P78A}-ctrl to image in vivo fluorescence dynamics while microinjecting PACAP₁₋₃₈ through nearby cannula (**Figure 4f**). Microinfusion of PACAP₁₋₃₈ (300 μ M, 200 nl) led to peak fluorescence increases of 126% $\Delta F/F_0$ in PAClight1_{P78A}-expressing mice. The fluorescence peaked at 28 min and dropped to 88% $\Delta F/F_0$ 1 hour after, when injections were positioned in average approximately 350 μ m from the recording site. In contrast, PAClight1_{P78A}-ctrl-expressing mice showed a fluorescence increase of only 16% $\Delta F/F_0$ (**Figure 4g**).

In conclusion, we show that PAClight1_{P78A} can be a useful tool to detect PACAP₁₋₃₈ in vivo with a dynamic range that allows for the detection of drug injections even in the presence of intact endogenous PACAP systems in mice. Moreover, our data suggest that PACAP₁₋₃₈ diffuses efficiently across hundreds of μ m, with slow extracellular degradation in neocortical brain areas of mice.

Two-photon validation of the sensor in living zebrafish

Zebrafish larvae are an important animal model that has long been recognized for its utility and applicability to drug discovery^{34–36}. Moreover, PAC1 signaling has been associated with the adaptive stress response of zebrafish^{10,37}. The high degree of amino acid sequence conservation of PACAP across the phylogenetic tree, as well as the *in vitro* response of PAClight1_{P78A} to zebrafish PACAP that we observed, motivated us to also functionally validate our sensor for use in live zebrafish larvae. Based on a publicly available single-cell RNA sequencing (scRNAseq) dataset³⁸, we identified the olfactory region of 4 day post-fertilization (dpf) old zebrafish larvae to express high levels of *Adcyap1b* in ~75% of cells composing the olfactory

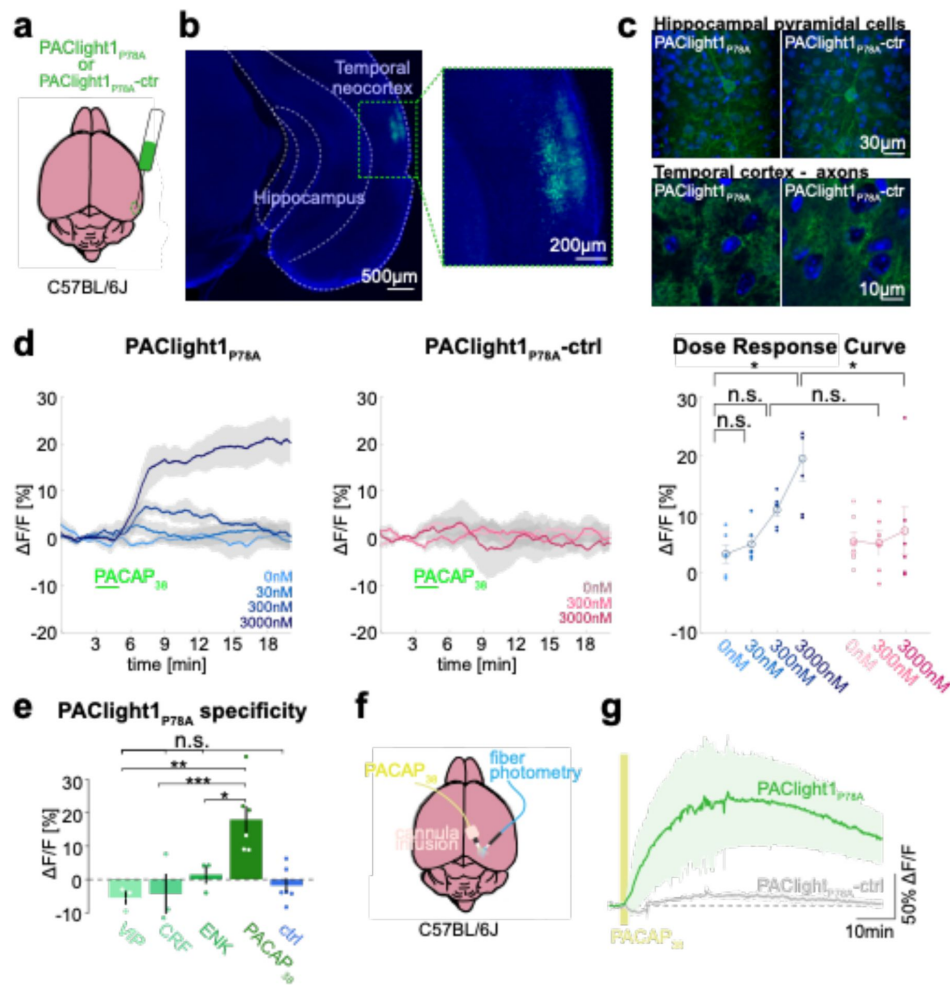


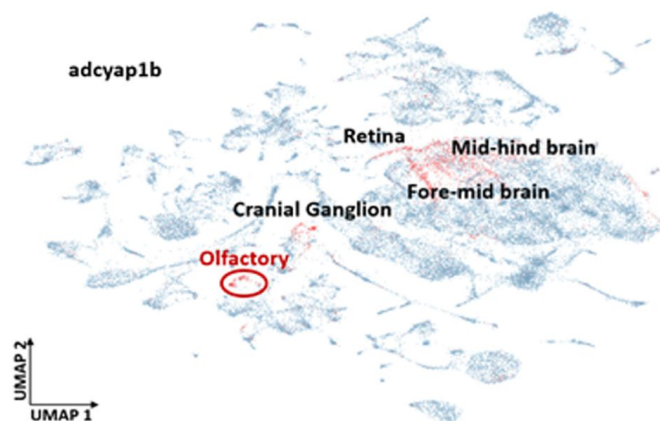
Figure 4

Ex vivo and in vivo sensor characterization in mice

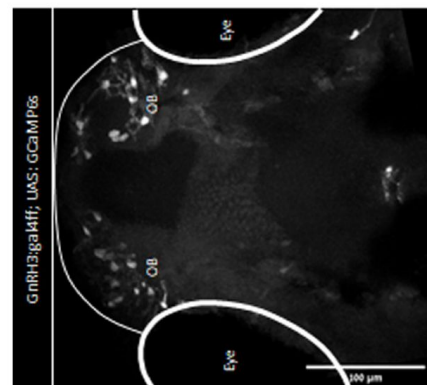
a) AAVs encoding the PAClight1_{p78A} or PAClight1_{p78A-ctrl} sensor were injected into the temporal neocortex of adult mice. **b)** Representative epifluorescent image of PAClight1_{p78A} fluorescence after 4 weeks of expression time. **c)** Maximum intensity projections of exemplary confocal images of PAClight1_{p78A} and PAClight1_{p78A-ctrl}-expressing neurons and neuropil in hippocampus and cortex enhanced with GFP-immunostaining and counterstained with DAPI (blue). **d)** Acute mouse brain slices expressing PAClight1_{p78A} (left) and PAClight1_{p78A-ctrl} (middle) were used to test the sensitivity of the sensor in the mammalian brain. PACAP₁₋₃₈ was bath-applied for 2min (green bar) at indicated concentrations. Data shown as mean $\pm$ standard error of the mean (SEM). $N=6$ slices per condition from ≥ 3 mice. Right: Dose-response curve for PAClight1_{p78A} $\Delta F/F_0$ (blue shades) and PAClight1_{p78A-ctrl} $\Delta F/F_0$ (pink shades) in response to indicated concentrations of PACAP₁₋₃₈. Mann-Whitney-U-tests with Bonferroni corrections revealed no statistically significant difference in mean PAClight1_{p78A} peak responses to bath-application of 30nM ($p=2.909$), and 300nM PACAP₁₋₃₈ ($p=0.0519$) compared to 0nM PACAP₁₋₃₈. We found a significant difference between 0nM and 3000nM PACAP₁₋₃₈ ($p=0.013$). Comparing mean peak $\Delta F/F_0$ PAClight1_{p78A} responses with PAClight1_{p78A-ctrl} revealed a significant difference when bath-applying 3000nM ($p=0.041$), but not for 300nM ($p=0.084$). **e)** Quantification of mean PAClight1_{p78A} peak responses to bath-application of 3 μ M VIP, CRF, ENK, PACAP₁₋₃₈ and negative control (ctrl) in acute mouse brain slices. $N=3-6$ slices per condition from ≥ 3 mice. No statistically significant difference ($F_{(4,10)}=5.15$) was found between ctrl and VIP ($p=0.9705$), ctrl and CRF ($p=0.9937$) and ctrl and ENK ($p=0.9726$). A statistically significant difference ($F_{(3,8)}=9.19$) was detected between PACAP₁₋₃₈ and VIP ($p=0.007$), PACAP₁₋₃₈ and CRF ($p=0.001$) and PACAP₁₋₃₈ and ENK ($p=0.045$). Statistically significant differences are indicated with an asterisk, non-significant differences with n.s. **f)** AAVs encoding the PAClight1_{p78A} or PAClight1_{p78A-ctrl} sensor were injected into the neocortex of adult mice. Fiberoptic cannulae and acute microinfusion cannulae were implanted nearby. **g)** PAClight1_{p78A} and PAClight1_{p78A-ctrl} fluorescence changes upon microinfusion of 300 μ M PACAP₁₋₃₈ (200nl) recorded with fiber photometry in freely behaving mice. $N=2$ mice each. Data shown as mean $\pm$ SEM.

region. The presence of *Adcyap1b* expression in the olfactory region was further confirmed with data obtained by Fransworth et al.³⁹ (Figure 5a). To induce expression of PAClight1_{p78A} in the olfactory region, we used the gal4 driver line Tg(GnRH3:gal4ff), which can strongly drive expression (e.g. of GCaMP6s) in the olfactory bulbs (Figure 5b). In conjunction with this gal4 driver line, we used Tol2-mediated integration of a UAS-promoted PAClight1_{p78A} construct. At 4dpf we immobilized the larvae with low-melting point agarose and performed two-photon volumetric imaging of the olfactory region for 5 consecutive 3D volumes as baseline. The immobilized zebrafish larvae were then placed onto a micromanipulator platform to inject 50nL of a 1mM PACAP₁₋₃₈ in saline solution or 50 nL of saline only (negative control) into the ventricular space (i.e. ICV injection). Subsequently, the larvae were re-imaged for 15 additional 3D volumes to record potential alterations in the pixel intensity values emitted by the PAClight1_{p78A} sensor (Figure 5c, 5c'). We observed a strong increase in PAClight1_{p78A} fluorescence already at ~120 seconds post-ICV injection with $\Delta F/F_0$ levels continuously rising until the end of the post-ICV injection recordings (Figure 5d, 5d'). Peak $\Delta F/F_0$ levels and the Area Under the Curve significantly differed between PACAP₁₋₃₈/saline and saline-only injected animals ($n_{\text{saline}} = 5$, $n_{\text{PACAP1-38}} = 5$, Kolmogorov-Smirnov test $p = 0.0079$, Figure 5e, 5e'). The condition median for peak $\Delta F/F_0$ levels was 70-fold larger in PACAP₁₋₃₈/saline injected animals (145.76% $\Delta F/F_0$) than in saline-only injected animals (2.06% $\Delta F/F_0$) (Figure 5e). However, we noticed considerable variability in the extent of the dynamic range displayed by different ROIs within the same and also between different zebrafishes (Figure 5d', 5e').

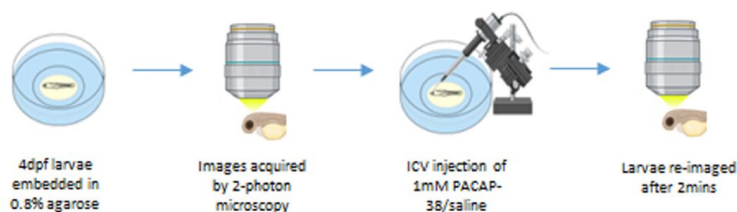
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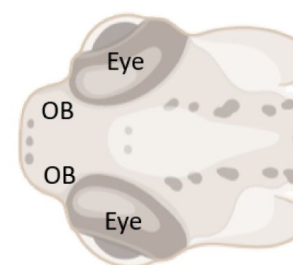
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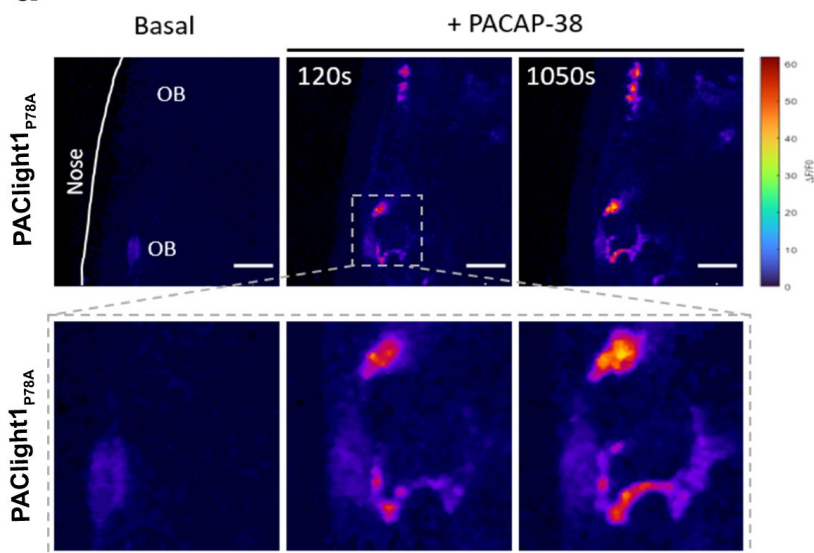
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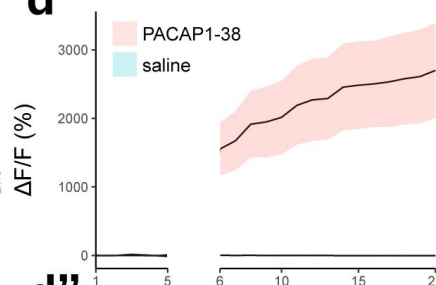
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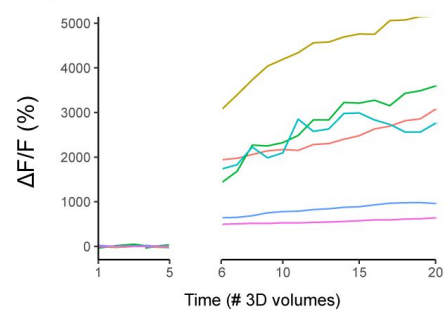
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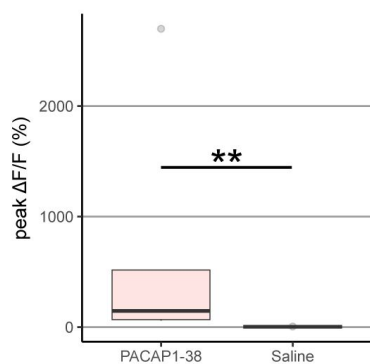
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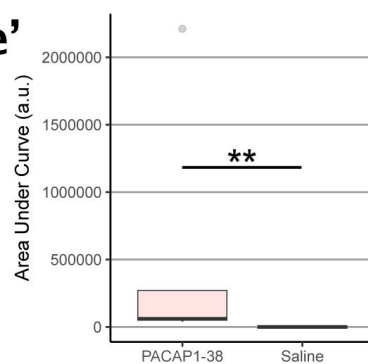


Figure 5

Characterization of PAClight1 in live zebrafish larvae

a) Uniform manifold approximation and projection (UMAP) showing topological distribution of single cell gene expression clusters with high *adcyap1b* (PACAP) expression of 0-4 dpf larvae. High *adcyap1b* expression is highlighted in the olfactory bulb. **b)** Maximum intensity projection image of the 4dpf *Tg(GnRH3:gal4ff; UAS:GCaMP6s)* larvae shows GnRH3 expressing cells in the olfactory bulb. **c)** Schema representing the experimental design. 4dpf larvae were immobilized and 3D volumetric images across time were obtained during the naïve state. 50nl of 1mM PACAP-38/saline was injected intracerebroventricularly and the same larvae was imaged again after 2mins. **c')** Schema of imaged region of interest including the olfactory bulb in the 4dpf larvae. **d)** (top) Representative maximum intensity projection of two-photon volumetric images showing naïve sensor at baseline and its increase in fluorescence after PACAP-38 injection in the same larvae along time (scale bar = 20µm). Colour bar is representative of the $\Delta F/F_0$ in the images. (bottom) Zoomed-in image of the left olfactory bulb showing increase in fluorescence after injection of PACAP₁₋₃₈. **d')** Representative quantification of the change in fluorescence with respect to basal fluorescence depicted as $\Delta F/F_0$. PACAP₁₋₃₈ injected larvae show an increase in activity of the fluorescent sensor as compared to their saline injected sibling controls. **d'')** Individual cell traces show variability in sensor responses in different cells that may be a function of expression of the sensor on the cell surface. **e,e')** Peak $\Delta F/F_0$ and Area under the Curve (AUC) of PACAP₁₋₃₈ injected larvae is significantly higher than the saline injected controls ($n_{\text{control}}=5$, $n_{\text{PACAP-38}}=5$, Kolmogorov-Smirnov test $p=0.0079$)

Discussion

In this work we engineered a new family of genetically-encoded fluorescent sensors using the human PAC1R as a GPCR scaffold. Two of the sensors that we developed (PAClight1 and PAClight1_{P78A}) exhibit a very high dynamic range (above 1000% $\Delta F/F_0$), excellent expression at the cell surface, high basal brightness, and retain the pharmacological profile and ligand-binding profile of the parent receptor, with the exception of PAClight1_{P78A} whose response to VIP is intentionally abolished via a single point mutation. As part of this sensor family, we also introduced a control sensor harboring the two point mutations R199A and Y161A, in which the response to PACAP₁₋₃₈ is abolished. Given the higher PACAP-selectivity of PAClight1_{P78A}, this sensor variant is intended to be employed when there is an experimental need to ensure maximal selectivity of the response for the endogenous ligand PACAP₁₋₃₈ (for example if the sensor is to be used for attempting the detection of endogenous PACAP₁₋₃₈ release). For all other experimental scenarios, PAClight1 can be a better choice, as it retains the original wild-type sequence in its ECD and thus most closely represents the natural ligand-binding profile of the human PAC1 receptor.

Importantly, we demonstrate successful expression and functionality of PAClight1_{P78A} and PAClight1_{P78A}-ctrl in the mouse brain through stereotactic injection of AAVs encoding the sensor. Additionally, our experiments in zebrafishes, showed intriguing inter- and intraindividual differences in maximal sensor response to ICV injection of the peptide ligand. These could potentially be explained by differences in the expression density of PAClight1_{P78A} within single cells. An intriguing alternative explanation – that receives support from the observation of continuously increasing $\Delta F/F_0$ levels – could be differential diffusion rates of PACAP₁₋₃₈ from the ventricular spaces into the brain parenchyma. Furthermore, there could be a dependence of the extent of the fluorescent response on the spatial distance between PAClight1_{P78A}-expressing cells and the ventricular wall. Under more controlled experimental conditions, PAClight1- expressing zebrafishes (or PAClight1_{P78A}, respectively) could therefore potentially be used to investigate and

validate diffusion properties of PAC1R ligands in live organisms at a relatively high-throughput or to screen for novel small-molecule or peptide PAC1R agonists and antagonists with improved penetration properties across the ventricular wall.

In line with the zebrafish experiments, PACAP₁₋₃₈ microinjections into mouse cortex revealed slow increases in $\Delta F/F_0$ levels even when PACAP₁₋₃₈ was injected only 100s of μm away from the recording site, substantiating the hypothesis of slow diffusion of PACAP₁₋₃₈ through brain tissue. These results show that the new tools introduced in this study can be valuable assets to investigate real-time dynamics of PAC1 receptor activation in response to specific PAC1 receptor agents at a cellular level and in vivo in zebrafish and in the mammalian brain. The in vivo validation of our sensor demonstrates its ability to reveal localized drug or peptide actions and characterize their diffusion dynamics within brain tissue. We therefore suggest that this tool can be employed in applications such as studying the diffusion of novel PAC1-targeting drugs and peptides to provide mechanistic insights into the localization of drug actions in the brain.

Notably, in this work we did not demonstrate the use of any of our PAClight sensors for the detection of endogenous PACAP release in tissues or animal models, as our focus was deliberately set on the characterization and validation of the tools for applications to drug screening and/or development across species. Future work could focus on employing these tools to test whether or not they could be suitable for detecting endogenous PACAP/VIP dynamics with high spatiotemporal resolution in tissues or awake behaving animals.

Methods

Molecular cloning

A synthetic DNA geneblock for the human PAC1R-null receptor sequence (PAC1R) was designed and ordered (Life Technologies) based on the NCBI protein data bank entry “NP_001109.2”. The protein-coding sequence was codon optimized and flanked by HindIII (5') and NotI (3') restriction sites for cloning into a pCMV plasmid (Addgene #60360). A hemagglutinin (HA) signal peptide (MKTIIALS YIFCLVFA) was introduced 5' to the PAC1R coding sequence. Circular Polymerase Extension Cloning (CPEC) was used to replace the intracellular loop 3 (Q336-G342) of PAC1R with a cpGFP module from dLight1¹¹. For sensor optimization, libraries of sensor variants were created through site-directed mutagenesis. For signaling assays, a linker + SmBit sequence⁴⁰ (GNSGSSGGGSGGGGSSGG + VTGWRLCERILA) was cloned at the 3' end of both the PAClight1_{P78A} and the PAC1R sequence. The XhoI cleavage site from within the SmBit linker sequence and the first 4 residues of the SmBit linker (GNSG) were first cloned into the 3' end of the PAClight1_{P78A} and PAC1R sequences using PCR. Subsequently, a restriction digest using XhoI and XbaI restriction enzymes was performed on the SmBit containing plasmid (B2AR-SmBit) as well as on the pCMV_PAClight1_{P78A} and pCMV_PAC1R plasmids. Ligation of the linker+SmBit insert into the linearized backbones was performed after gel extraction and purification of the insert. For generation of stably expressing and inducible Flp-In T-REx 293 cells, the PAClight1_{P78A} and PAClight1 sequences were cloned into the pcDNA5/FRT/TO backbone, respectively, using restriction digest with BamHI and NotI restriction enzymes. Agonist and antagonist plasmids (i.e. PURE_FLAG-TEV-PACAP1-38 and PURE_FLAG-TEV_PACAP6-38 respectively) used for validation of the microfluidics platform were obtained by cloning the nucleotide sequences for PACAP₁₋₃₈ and PACAP₆₋₃₈ (obtained as synthetic geneblocks) into the PURE_deGFP backbone using Gibson Assembly. Subsequently, a FLAG-tag (DYKDDDDK) as well as the first 5 residues of the recognition sequence for the TEV protease (ENLYFQ) were cloned upstream of the PACAP sequences into the obtained plasmid. Single colonies were picked and sequence verified. PCR reactions were performed using a Pfu-Ultra II Fusion High Fidelity DNA Polymerase (Agilent), whereas Gibson Assembly was performed using NEBuilder HiFi DNA Assembly Master Mix (New England Biolabs). All sequences were verified using Sanger sequencing (Microsynth).

Structural modelling, protein sequence alignment and peptide synthesis

Modelling of protein structure for the PAClight1_{P78A} sensor construct was performed using AlphaFold2⁴¹, using pdb70 as a template mode. The best-scoring prediction was then manually edited using UCSF Chimera (version 1.13.1). Multiple sequence alignment was performed using ClustalOmega⁴² and visualized with the Jalview⁴³ software (version 2). Zebrafish PACAP2₁₋₂₇(zfPACAP2), chicken PACAP₁₋₃₈ (chPACAP38), amnesiac, maxadilan (Maxadilan₁₋₆₁), and Max.d.4 peptides were synthesized on an automated fast-flow peptide synthesizer (AFPS) using a previously described protocol⁴⁴.

Cell culture, imaging and quantification

HEK293T cells (ATCC #CRL-3216) were cultured in DMEM medium (Thermo Fisher) supplemented with 10% FBS (Thermo Fisher) and 1x Antibiotic-Antimycotic (100 units/mL of penicillin, 100µg/mL of streptomycin, and 0.25 µg/mL of Amphotericin B, Thermo Fisher) and incubated at 37°C with 5% CO₂. Cells were transfected at 50-60% confluency in glass-bottomed dishes using the Effectene transfection kit (Qiagen) according to manufacturer instructions, and imaged 24-48h after transfection, as previously described¹⁷. Primary cultured hippocampal neurons were prepared and transduced as previously described¹⁹. Before imaging, all cells were rinsed with 1mL of Hank's Balanced Salt Solution (HBSS, Life Technologies) supplemented with Ca²⁺ (2mM) and Mg²⁺ (1mM). Time-lapse imaging was performed at room temperature (22° C) on an inverted Zeiss LSM 800 confocal microscope using either a 40x oil-based or a 20x air objective. Longer-term imaging for determination of internalization/stability as well as reversibility was performed using a 10x air objective, with 1 frame (1024x1024 pixels) acquired every minute (2x pixel averaging). Imaging was performed using a 488nm laser as excitation light source for PAClight1 sensors. During imaging, ligands were added in bolus on the cells using a micropipette to reach the final specified concentrations of ligands on the cells. For quantification of recordings in single dishes, regions of interest (ROI) were selected manually using pixel-intensity based thresholding in Fiji to isolate the cell membrane. Sensor response ($\Delta F/F_0$) was calculated as the following: $(F(t)-F_{base})/F_{base}$ with $F(t)$ being the ROI fluorescence value at each time point (t), and F_{base} being the mean fluorescence of the 10 time points prior to ligand addition.

Stable cell line generation and maintenance

The stable cell line for tetracycline-inducible expression of PAClight1_{P78A} was generated following previously described procedures²⁹. Flp-In T-REx 293 cells were grown in DMEM supplemented with 10% (v/v) fetal bovine serum (FBS; Invitrogen), 100µg/mL zeocin (ThermoFisher Scientific), and 15µg/mL blasticidin (ThermoFisher Scientific). To obtain a cell line with stably-integrated PAClight1_{P78A} or PAClight1 expression cassettes, cells were grown in T150 flasks (Corning) until 70% confluency, were then transfected with 0.6µg of pcDNA5/FRT/TO- PAClight1_{P78A} DNA vector and 5.4µg pOG44 vector using Effectene transfection kit (Qiagen). Two days after transfection, cells were split and the medium was changed to DMEM supplemented with 10% (v/v) FBS, 200µg/mL Hygromycin B (Sigma), and 15µg/mL blasticidin. The medium was then replaced twice a week until individual colonies were visible. An individual colony was manually selected and expanded for subsequent experiments. Induction of PAClight1_{P78A} or PAClight1 expression was obtained by adding 1µg/mL doxycycline (Sigma) to the cell medium 1- 2 days prior to experimentation.

Spectral characterization of the sensors

One-photon spectral characterization of the PAClight1_{P78A} sensor was performed using PAClight1_{P78A}-transfected HEK293T cells before and after addition of PACAP₁₋₃₈ (10µM). One-photon fluorescence excitation (λ_{em} = 560nm) and emission (λ_{exc} = 470nm) spectra were determined on a Tecan M200 Pro plate reader at 37°C. 24 hours after cell transfection in 6-well

format (linear PEI), ~1 million cells were dissociated with addition of TrypLE Express (ThermoFisher) and thoroughly washed with PBS. Next, cells were resuspended in 300µl of PBS and aliquoted into two individual wells of a 96-well microplate with or without PACAP₁₋₃₈ (10µM), together with two wells containing the same amount of non-transfected cells to account for autofluorescence and a single well containing PBS for subtraction of the Raman bands of the solvent.

Flow cytometry

HEK293T cells and stable HEK293_PAClight1_{p78A}/PAClight1 cells were seeded into T175 flasks and grown to 50-60% confluency under culture conditions described above. HEK293T cells were then transfected with 20µg of pCMV_PAClight1_{p78A} or pCMV_PAClight1 using linear PEI (Sigma-Aldrich; #764965) and a PEI-to-plasmid ratio of 3:1. Stable HEK293_PAClight1_{p78A}/PAClight1 cells were induced with 1µg/mL doxycycline (Sigma) at 50- 60% confluency. Two days after transfection or induction, cells were washed in 1x PBS before detachment with TrypLE Express (ThermoFisher) and subsequent centrifugation at 150xG for 3 minutes. Palleted cells were then resuspended in ice-cold FACS buffer containing 1xPBS, 1mM EDTA, 25mM HEPES (pH 7.0), 1% fetal bovine serum and diluted to $4-4.8 \times 10^6$ cells/mL. A dilution series of PACAP₁₋₈ and VIP was prepared in FACS buffer. 125µL of peptide solution was pipetted into a 96-well format and another 125ul of cell suspension was added to each well. The 96-well U-bottom plate was then loaded into a BD FACS Canto II cytometer equipped with a high- throughput sampler for 96-well format sampling. Voltage for the photomultiplier tubes was set to 200V for the forward scatter detector, to 400V for the side scatter detector, and to 350V for the FITC channel fluorescence detector. Excitation was performed at 488nm, while emission was directed through a 502LP mirror and a 530/30 band pass filter. Sampling from 96-well format wells was performed with an initial mixing step (3x 100µl mixes at 150µL/s) followed by sample acquisition of 200µL at a flow rate of 1-3µL/s. Sampling was performed until 100K events were recorded per well. After sample acquisition of each well, a 800µL wash step of the sampling line was performed. Sample acquisition within a dilution series replicate was performed from low-to- high concentrations to minimize potential peptide carry-over into the neighboring wells. Furthermore, a minimum of six wells without PACAP or VIP present were sampled between each replicate of the same construct. Raw data was exported in FCS3.1 format and further processed for analysis within R. The following R packages were used for the analysis and visualization of the flow cytometry experiments: flowCore, flowAI, ggcyto, dplyr, tidyr, forcats, purrr, stringr, openCyto, openxlsx, svglite, ggprism, ggnewscale, and glue. After import of the raw datasets, a rectangular gate (“Cells”) was defined on a FSC-A vs. SSC-A scatter plot with the limits being set to 25K-infinity and 15K-infinity, respectively. Next, a “Singlet” gate within the parent gate “Cells” was created automatically using the openCyto::singletGate function based on the FSC-A vs. FSC- H scatter plot. Median fluorescence intensities (MFI) of the FITC-A channel for all recorded events within the “Singlet” gate were calculated for each well. The MFI values were then normalized (“NormFITC_A”) within each titration replicate to the average of the 0nM peptide condition. Dose- response-curves and EC₅₀ values were obtained by first grouping datasets by construct and peptide conditions. For each group, a non-linear least squares model was fit using the following formula:

$$NormFITC_A \sim \min + ((\max - \min) / (1 + \exp(hill_coefficient * (EC50 - Conc_M))))$$

For the VIP titration on the non-responsive sensor variant, a linear regression model was fit using the following formula:

$$NormFITC_A \sim Conc_M$$

NanoLuciferase complementation assays

HEK293 cells were seeded in 6-well plates and transfected with 0.25 µg LgBiT-miniG (miniGs or miniGsq⁴⁵) or LgBiT-β-arrestin2⁴⁰ and 0.25 µg SmBiT-tagged receptor plasmids using 3 µL Lipofectamine 2000 (Thermo Fisher). 24 hours after transfection, cells were plated onto black clear-bottomed 96-well plates at 50,000 to 100,000 cells/well or 384-well plates at 20,000 to 30,000 cells/well in DMEM without phenol red, supplemented with 30 mM HEPES (pH 7.4) and Nano-Glo Live Cell Reagent (Promega), and incubated for 45 min at 37°C. Luminescence signal was measured simultaneously across the plate using the FDSS/µCELL plate reader (Hamamatsu). Baseline luminescence (before agonist addition) was acquired for 3 min. Vehicle (buffer) or agonist (PACAP₁₋₃₈, 1 µM) were added simultaneously into wells by an integrated dispensing unit. Luminescence was recorded every 1–2 s for 8 min post agonist addition. Agonist-treated wells were initially normalized to vehicle wells, luminescence intensity was then normalized to the baseline prior to ligand addition.

Generation and imaging of transgenic fishes

The PAClight1_{p78A} sequence was cloned into a Tol2 transposase-based plasmid⁴⁶ with a UAS promoter. The plasmid (75 ng/ul) was injected into *Tg(GnRH3:gal4ff)*⁴⁷ embryos at the single-cell stage along with transposase mRNA (25 ng/ul). Embryos were sorted at 3 days post fertilization using the red heart marker (cmc:mCherry) in the *UAS:PAClight1_{p78A}* construct. Positive zebrafish larvae were immobilized using alpha-bungarotoxin (1 mg/ml for 30 s) and mounted ventrally (ventral side down) on to a custom molded 3 cm plate using 0.8% low-melt agarose (Sigma-Aldrich). Plates were perfused with E3 media (5 mM NaCl, 0.17 mM KCl, 0.33 mM CaCl₂, 0.33 mM MgSO₄ in H₂O) prior to imaging. Larvae were imaged under the Leica TCS SP8 Multi-Photon Microscope with a 25X/0.95NA water immersion objective. Volumetric images across time were obtained after irradiation at 890 nm at 400 Hz/frame with 2x averaging. Larvae were imaged for five volumes in the naïve state followed by intracerebroventricular injections (ICV) of 1 mM PACAP-38 (Sigma-Aldrich) in saline solution (0.6% NaCl, 0.02% Na₂CO₃) and only saline for controls. Larvae were then re-imaged approximately 120 seconds after ICV injection using the same imaging parameters for 15 volumes. Basal and post-injection images were concatenated and registered using descriptor-based series registration on ImageJ/Fiji (https://github.com/fiji/Descriptor_based_registration.git). Raw fluorescence traces were extracted from these images from manually drawn ROIs around cells. Statistical analysis and plotting of graphs were performed in MATLAB and R. Baseline fluorescence, F_0 , was computed using basal fluorescence intensities, which were then used to calculate $\Delta F/F_0$. The Kolmogorov-Smirnov t-test was performed where applicable. Single cell transcriptomic data³⁹ from 4 dpf zebrafish larvae was plotted as a function of UMAPs in the UCSC cell browser. High expression of *adcyap1b* (PACAP) was visualized in Cluster 94 (<https://cells.ucsc.edu/?ds=;zebrafish-dev&gene=;ENS DARG0000027740>). *Tg(GnRH3:gal4ff)* transgenic line was used to express the PAClight1_{p78A} in the olfactory bulb (OB). To confirm that GnRH3 cells were indeed present in the OB at 4 dpf, a *UAS:GCaMP6s* construct was expressed in the *Tg(GnRH3:gal4ff)* line.

Virus production

The biosensor AAV constructs were cloned in the Patriarchi laboratory, while the opsin AAV construct was constructed by the Viral Vector Facility of the University of Zürich (VVF). All viral vectors were produced by the VVF. The viral titers of the viruses used in this study were: AAV9.hSyn.PAClight1_{p78A}, 0.75–1.5 × 10¹³ GC/mL, AAV9.hSyn.PAClight1_{p78A}-ctrl, 1.6 × 10¹³ GC/mL.

Animals

Rat embryos (E17) obtained from timed-pregnant Wistar rats (Envigo) were used for preparing primary hippocampal neuronal cultures. Wild-type C57BL/6J mice (Janvier, 6–10 weeks old) of both sexes were used in this study. Mice were kept in standard enriched cages with *ad libitum*

access to chow and water on either normal or reversed 12-h/12-h light/dark cycle. Mice were housed in cages of 2-5 animals.

All procedures in mice were performed in accordance with the guidelines of Medical University of Vienna and under approved licenses by the Austrian Ministry of Science.

Zebrafish were maintained and bred by standard protocols and according to FELASA guide- lines. All experiments using zebrafish were approved by the Weizmann Institute's Institutional Animal Care and Use Committee (Application number 06340722-1).

Stereotactic surgeries

For ex vivo slice imaging and validation of mouse brain expression, 6 to 10-week-old male and female C57BL/6J mice (Janvier) were anesthetized with 5% isoflurane and maintained under stable anesthesia during the surgery (1.3-2% isoflurane). Lidocaine (3.5mg/kg; Gebro Pharma, #100562 1404) and Carprofen (4mg/kg; Zoetis, #256684) were administered subcutaneously for local anesthesia and general analgesia. A small craniotomy was made bilaterally, targeting the temporal neocortex (Injection site at -4.1 mm posterior to bregma, as lateral as possible and 1.2 mm ventral to the pial surface). A glass micropipette was used for virus delivery and inserted slowly into the brain. Before and after virus delivery, the micropipette was kept in place for a waiting time of 8 min. 200nL of AAV9.hSyn.PAClight1_{P78A} or AAV9.hSyn.PAClight1_{P78A}-ctrl were injected, with an injection rate of 40nL/min using a microsyringe pump (KD Scientific; #788110). The pipette was withdrawn slowly and the skin of the skull was sutured. The mice were allowed to recover for 4 weeks before being used for the experiments, with Carprofen (4 mg/kg) being administered on the two consecutive days after the surgery.

Slice imaging

The mice were deeply anesthetized with isoflurane before being sacrificed by a transcardial perfusion with 30mL cooled sucrose solution (212mM sucrose, 3mM KCl, 1.25mM NaH₂PO₄, 26mM NaHCO₃, 1mM MgCl₂, 0.2mM CaCl₂, 10mM glucose) oxygenated with carbogen gas. The mice were decapitated and the brain was dissected out. 300μM coronal slices were prepared using a Leica VT1200 vibratome in cold oxygenated sucrose solution (0.12 mm/s, 0.8 mm amplitude). The slices were incubated in oxygenated Ringer's solution (125mM NaCl, 25mM NaHCO₃, 1.25mM NaH₂PO₄, 2.5mM KCl, 2mM CaCl₂, 1mM MgCl₂, 25mM glucose) at 37°C for 10min and afterwards kept at room temperature. The slices were imaged using an Olympus BX51WI microscope with an ORCA-Fusion Digital CMOS camera (Hamamatsu, #C14440-20UP) and a 10x objective (UMPLFLN10XW objective, Olympus). An EGFP filterset was used to control emission and excitation spectra (470/40, 525/50; AFH #F46-002). Micromanager software ⁴⁸ was used to synchronize the excitation LED with the camera shutter. Videos were taken at 4 Hz framerate using 100 ms exposure time per frame. Cortical areas with clearly visible baseline fluorescence were selected for imaging. Baseline recordings of 10-20 min were performed prior to data acquisition to allow the slices to accustom to the conditions. Bath temperature was set to ~31°C. The experimental recording was 20 min long, with the respective peptide being infused from minute 3-5. The following peptides were used: PACAP₁₋₃₈ (30-3000nM; Phoenix Pharmaceuticals; #052-05), VIP (3000nM; Phoenix Pharmaceuticals; #064-16), CRF (3000nM; Phoenix Pharmaceuticals; #019-06), and Met-Enkephalin (3000nM; Phoenix Pharmaceuticals; #024-35). Of note, since our preliminary data suggested a decrease in PAClight1_{P78A} fluorescence in response to 3 μM CRF, we enriched the Ringer's solution with synaptic transmission and neuronal activity blockers (NBQX, CPP, CGP 55845, Gabazine, and TTX) to prevent recruitment of neuronal circuits that could lead to endogenous PACAP₁₋₃₈ release (Tocris, biotechnie; #Tocris 1262/50, #Tocris 0373/50, # Tocris0247/50, #Tocris 1248/50, #Tocris 1069/1).

Data analysis was performed using a custom-written Matlab script. Movement correction as well as bleaching correction was performed. The average fluorescence across the whole imaging window was normalized to the baseline fluorescence measured during the first 3 min of the recording prior to peptide infusion to calculate $\Delta F/F_0$: $(F(t)-F_{base})/F_{base}$ with $F(t)$ being the fluorescence value at each time point (t), and F_{base} being the mean fluorescence of the first 3 min. 6 slices from at least 3 different mice were averaged for calculating the dose-response curve. For PAClight1_{p78A} responses to VIP, CRF and ENK, 3 slices from 3 different mice were averaged.

In vivo photometric imaging and microinfusion

Craniotomies were made as described for stereotactic injections. Tapered fiberoptic cannula implants (MFC_200/230-0.37_2mm_MF1.25_A45) with low autofluorescence epoxy were implanted into the right neocortex at 0.9 mm depth with the angled (uncoated) side of the fiber tip facing towards the left. The implant was stabilized with cyanoacrylate glue.

Cannulae were surgically implanted to locally infuse PACAP₁₋₃₈ into the cortex. Stainless steel guide cannulae (26 gauge; C315GA/SPC, InVivo One, Roanoke, VA, USA) were positioned in cortical L1, such that the tip of the internal cannula terminated 400 μ m medial to the fiberoptic cannula. The implant was secured with cyanoacrylate glue. Dummy cannulae that did not extend beyond the guide cannulae (C315DC/SPC, InVivo One) were inserted to prevent clogging.

Mice were handled for 7 days prior to drug infusions

For photometric recordings, a 200 μ m diameter and 0.37 NA patchcord (MFP_200/220/900-0.37_5m_FCM-MF1.25, low autofluorescence epoxy, Dorics) was used to connect the implanted fiberoptic cannulae to a Dorics filter cube for blue (465–480 nm) excitation light. Emission light was detected by built-in photodetectors (500–540 nm). Signals from the photodetectors were amplified with Dorics built-in amplifiers and acquired using a Labjack (T7). LJM Library (2018 release) was used to allow communication between MATLAB and Labjack. The voltage output from the LED drivers was amplitude modulated at 171 Hz (sine wave) as described previously ⁴⁹. Amplitude modulation was programmed in MATLAB. 470 nm LEDs (M470F3, Thorlabs; LED driver LEDD1B, Thorlabs) were used. Light power at the patchcord tip was set to an average of 45 μ W. Fluorescence was calculated with custom-written MATLAB scripts based on a previous publication (Owen and Kreitzer, 2019). Photometry data were sampled at 2052 Hz.

For drug infusions, dummy cannulae were replaced by internal cannulae (33 gauge; C315LI/SPC, InVivo One) that extended 1 mm beyond the guide cannulae and were connected to a microinfusion pump. After a recovery time of 5–10 min, PACAP₁₋₃₈ (200 nM; 300 μ M diluted in normal Ringer solution, NRR) was infused at 100 nL/min (NRR in mM: 135 NaCl, 5.4 KCl, 5 HEPES, 1.8 CaCl₂, pH 7.2 adjusted with KOH, sterile-filtered with 0.2 μ m pore size).

Histology

To verify expression of the sensor in the mouse cortex and hippocampus, virus injections were performed with stereotactic surgeries as described above. The virus was allowed to express for four weeks, before the mice were transcardially perfused with PBS and 4% paraformaldehyde, before the brain was dissected and processed into 50 μ m coronal slices using a Leica VT1000S vibratome. Slices were immunostained with chicken anti-GFP antibodies (Abcam Cat# ab13970, RRID:AB_300798; 1:500 diluted). In brief, slices were permeabilized and blocked for 1 h with PBS containing 5% NGS and 0.2% Triton X-100, followed by a 24 h antibody incubation at 4°C in PBS containing 5% NGS and 0.2% Triton X-100. Sections were washed in PBS and then incubated for 1 h in 1:500 diluted goat anti-chicken IgG (H+L) cross-adsorbed secondary antibody (Alexa fluor 488; Fisher Scientific A11039). The mounted slices were imaged using a Leica SP8 X confocal microscope equipped with a 63x 1.3 NA glycerol immersion objective.

Statistical analysis

For pairwise analysis of sensor variants, the statistical significance of their responses was determined on a case-by-case basis using a two-tailed unpaired Student's t-test with Welch's correction. Analysis of variance (ANOVA) testing was followed by pairwise comparison with correction for multiple comparison with either Dunnett's correction or Hochberg correction. For data not meeting assumptions of normality or homoscedasticity, Mann-Whitney-U tests followed by Bonferroni corrections were applied. All p-values are indicated either in the results section or in the figure legends. Data of sensor screening and validation experiments is displayed as mean $\pm$ 1 standard deviation. No statistical methods were used to predetermine sample size in cultured cells. Power calculations were performed to determine sample size for experiments using mice.

Author contributions

TP led the study. TP and RBC conceived the development of the PAClight sensors. RBC performed all *in vitro* molecular cloning and screening that led to the development of PAClight sensors and characterized the optical and pharmacological properties of the sensor under the supervision of TP. AG measured and analyzed sensor kinetics under the supervision of TP. EW and PH synthesized peptides under supervision of NH. KA performed intracellular signaling assays under the supervision of MS. MAB prepared cultured neurons under the supervision of DB. SN and SM performed and analyzed imaging experiments in acute brain slices and *in vivo* photometry recordings and microinfusions under the supervision of SM. PR performed the *in vivo* experiments using zebrafish under the supervision of GL. TP and RBC wrote the manuscript with contributions from all authors.

Data availability

The DNA and protein sequence of the sensor developed herein have been deposited on NCBI (accession number OQ366523) and are available in Supplementary Data S1. Viral DNA plasmids have been deposited both on ADDGENE and on the UZH Viral Vector Facility (<https://vvf.ethz.ch/>). Viral vectors can either be obtained from the Patriarchi laboratory, the UZH Viral Vector Facility, or Addgene. Data presented in this manuscript are available from the authors upon reasonable request.

Code availability

Custom written code can be made available upon request.

Acknowledgements

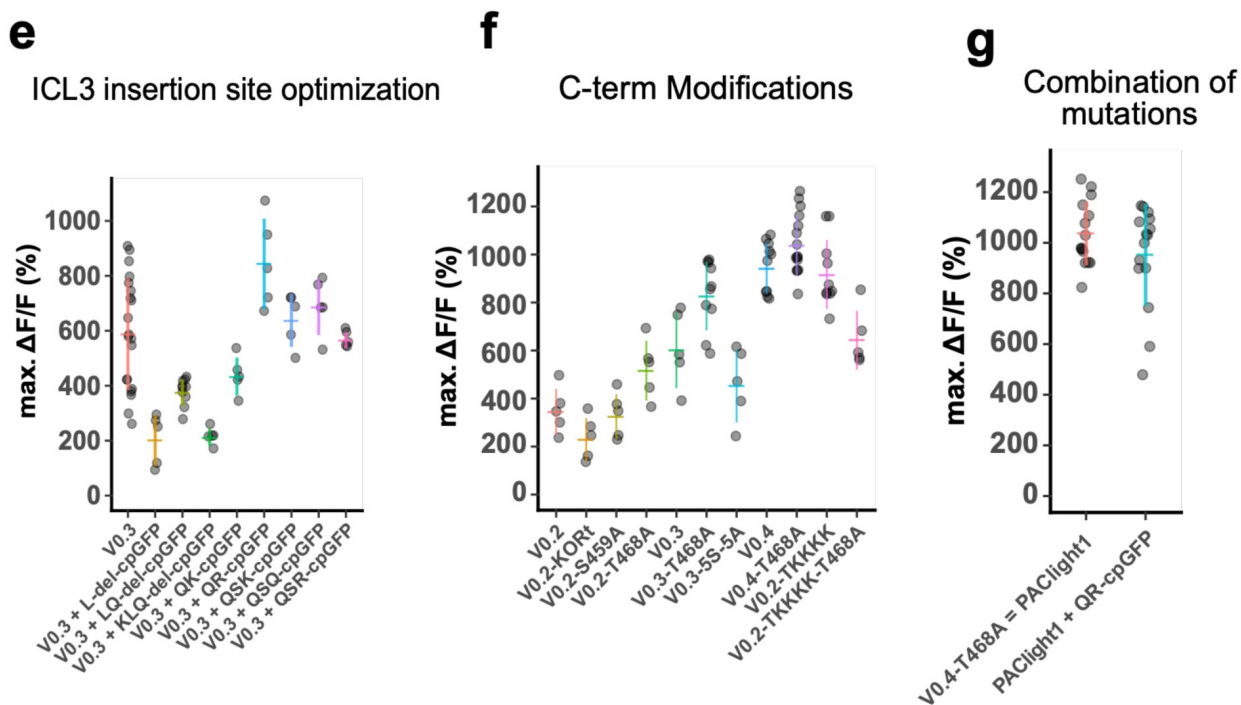
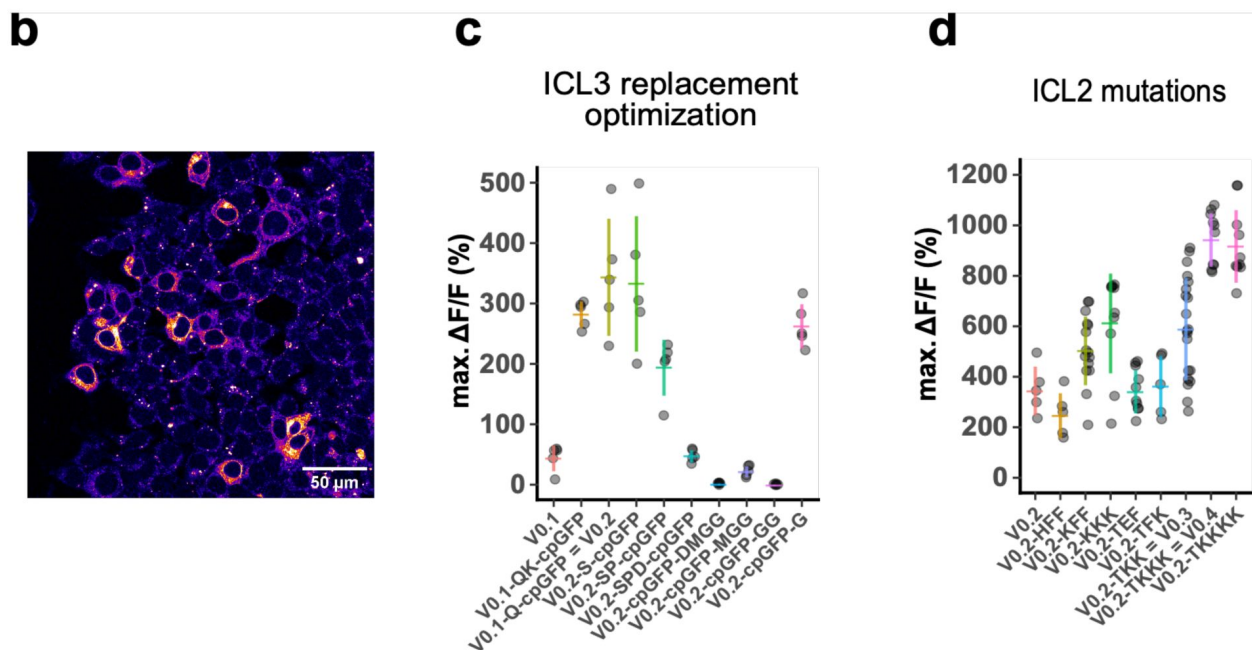
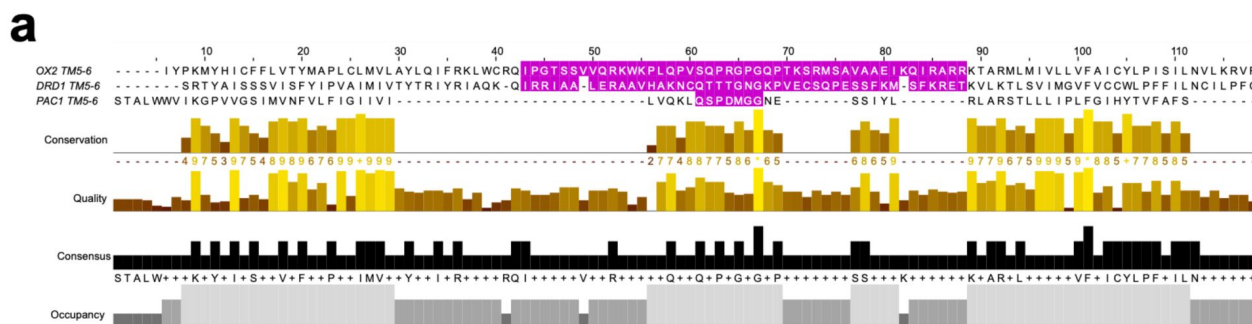
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Competing financial interests

TP is a co-inventor on a patent application related to the technology described in this article.

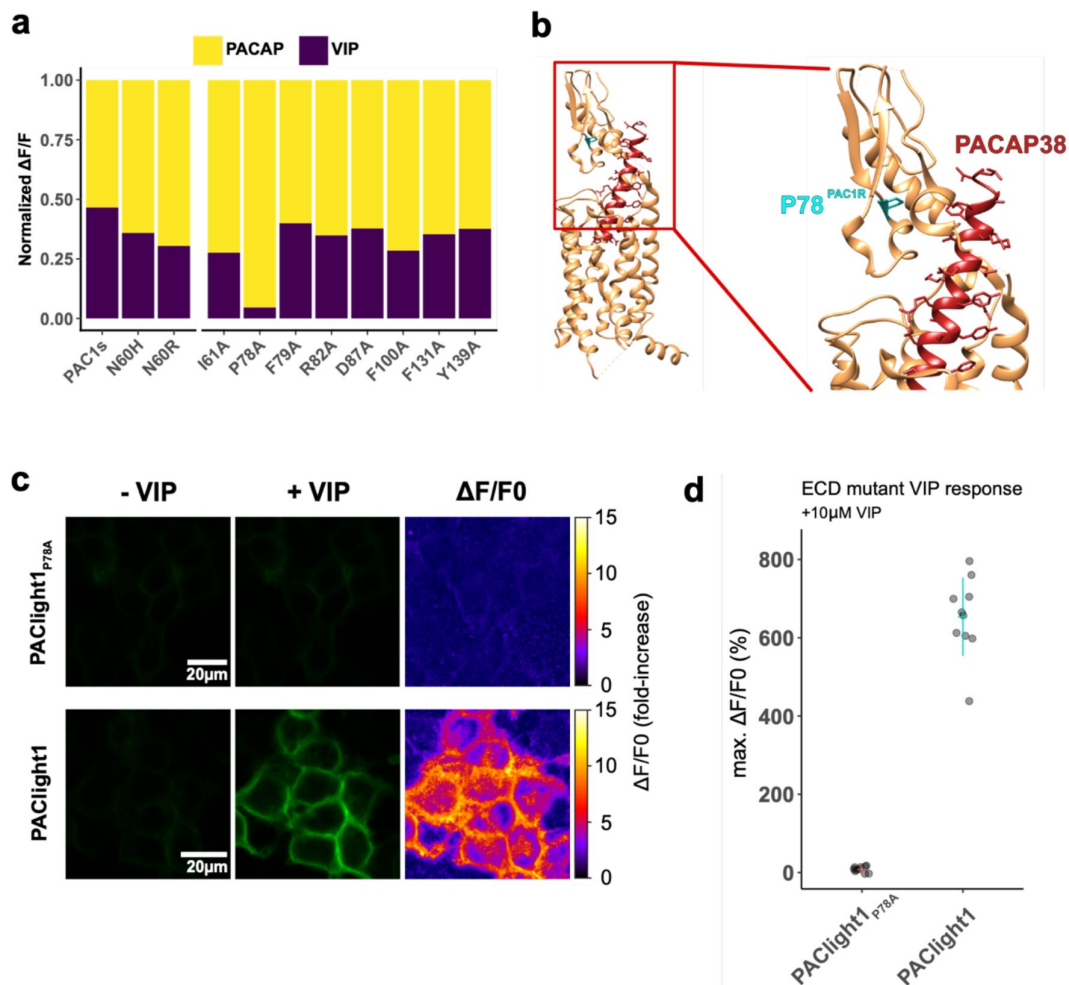
All other authors have nothing to disclose.



Extended Data Figure 1

Optimizing the fluorescent response of PAClight sensors

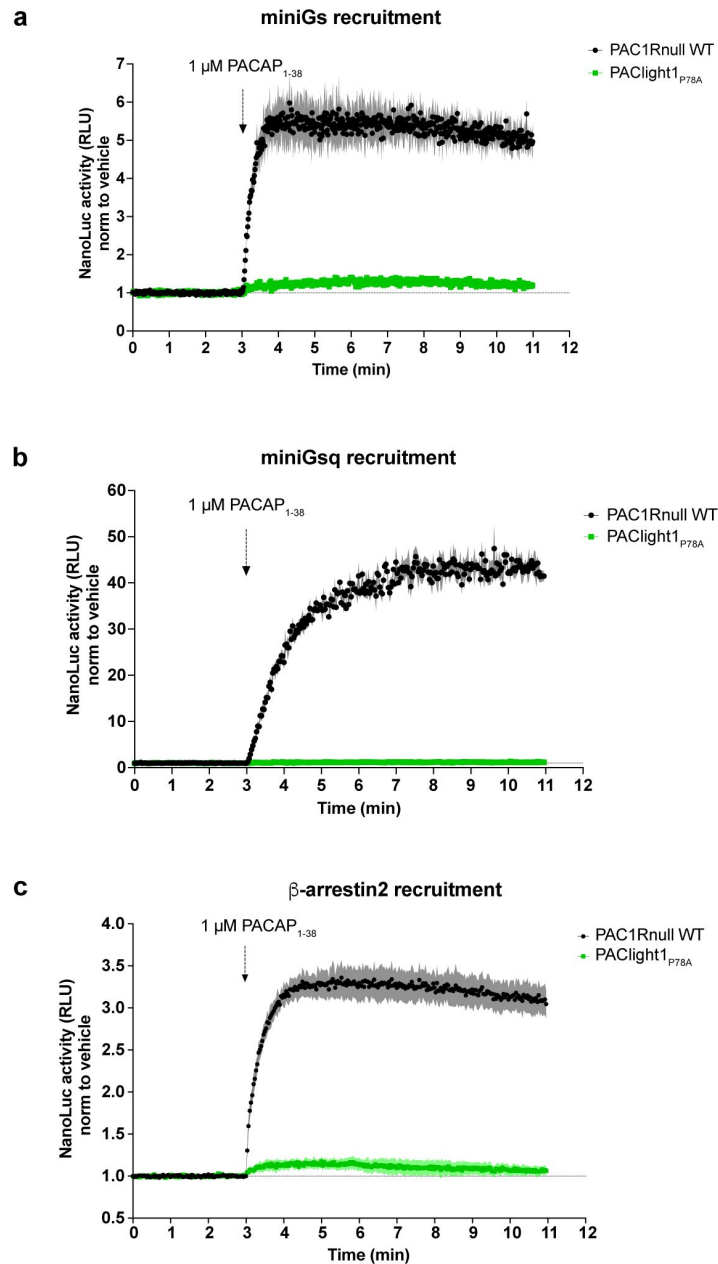
a) Multiple amino acid sequence alignment of the TM5, ICL3, and TM6 domains of the human orexin 2 receptor (OX2), the human D1 dopamine receptor (DRD1), and of the human PAC1Rnull (PAC1). The sequences replaced by the cpGFP module in the OxLight1, dLight1.3b, and PAClight1_{P78A} sensors are highlighted in magenta. The OX2 and DRD1 receptor sequences align well due to the fact that they both belong to the class A GPCRs. The PAC1Rnull belongs to the class B1 family of GPCRs and hence aligns less well to the OX2 and DRD1 backbones. We therefore guided the initial design and site of cpGFP module insertion by protein structure rather than by amino acid sequence. **b)** Pseudo-colored confocal micrograph of HEK293T cells expressing the prototypical PAClight_V0.1. Plasma membrane expression can be detected, but a significant amount of sensor protein is retained intracellularly. **c)** Scatterplot representing insertion site variants screened for maximum dynamic range. Reinsertion of the glutamine on the N-terminal side of the cpGFP module (PAClight_V0.2) drastically increased the dynamic range to 343% $\Delta F/F_0$ (compared to 43.4% $\Delta F/F_0$ in PAClight_V0.1). **d)** Point mutations in the intracellular loop 3 (ICL3) further improved the dynamic range to 588% $\Delta F/F_0$ for PAClight_V0.3 (point mutations: F259K/F260K) and to 942% $\Delta F/F_0$ for PAClight_V0.4 (point mutations: F259K/F260K/P261K). **e)** On the backbone of the improved ICL2 variant PAClight_V0.3 we screened again for the optimal insertion site of the cpGFP module. To this end we performed serial deletions as well as insertions of charged residues on the residues n-terminal to the cpGFP module. This small-scale screen identified a mutant with a glutamine and arginine insertion before the cpGFP module (PAClight_V0.3+QR-cpGFP) with improved dynamic range ($\Delta F/F_0 = 844\%$) compared to the PAClight_V0.3 backbone. **f)** To abolish β -Arrestin recruitment and further improve membrane localization, we further screened C-terminus mutants. We included mutants in this screen in which we replaced the C-terminus with the tail of the kappa opioid receptor (KOR_T), as well as point mutants of known post-translational modification sites and serine residues. The point mutation of the T468 to alanine further improved plasma membrane localization (data not shown) and improved the dynamic range from 942% $\Delta F/F_0$ (in PAClight_V0.4) to 1037% $\Delta F/F_0$. The resulting construct was named PAClight1. **g)** Since the insertion of QR in front of the cpGFP module had a positive effect on dynamic range on the backbone of PAClight_V0.3 (**e**), we also tested this insertion of QR on the PAClight1 backbone. However, the QR insertion did not yield improved dynamic range on the PAClight1 backbone ($\Delta F/F_0 = 953\%$) and was hence rejected. Each datapoint represents the average of the max $\Delta F/F_0$ within one ROI. In each replicate 5 ROIs each were quantified. $N=4$ for V0.3. $n=3$ for V0.2_KFF, PAClight1, and PAClight1+QR-cpGFP. $n=2$ for V0.2_KKK, V0.2_TEF, V0.2_KKKK, V0.3_LQ-del-cpGFP, V0.3_T468A, and V0.4. For all other constructs $n=1$ replicates each was screened. The y-axis position of the colored horizontal bars represents the mean across all replicates within a construct. The extent of the colored vertical bars represents ± 1 standard deviation. For ease of comparability some constructs are potted on multiple plots.



Extended Data Figure 2

Engineering PACAP selectivity in PAClight1_{P78A}

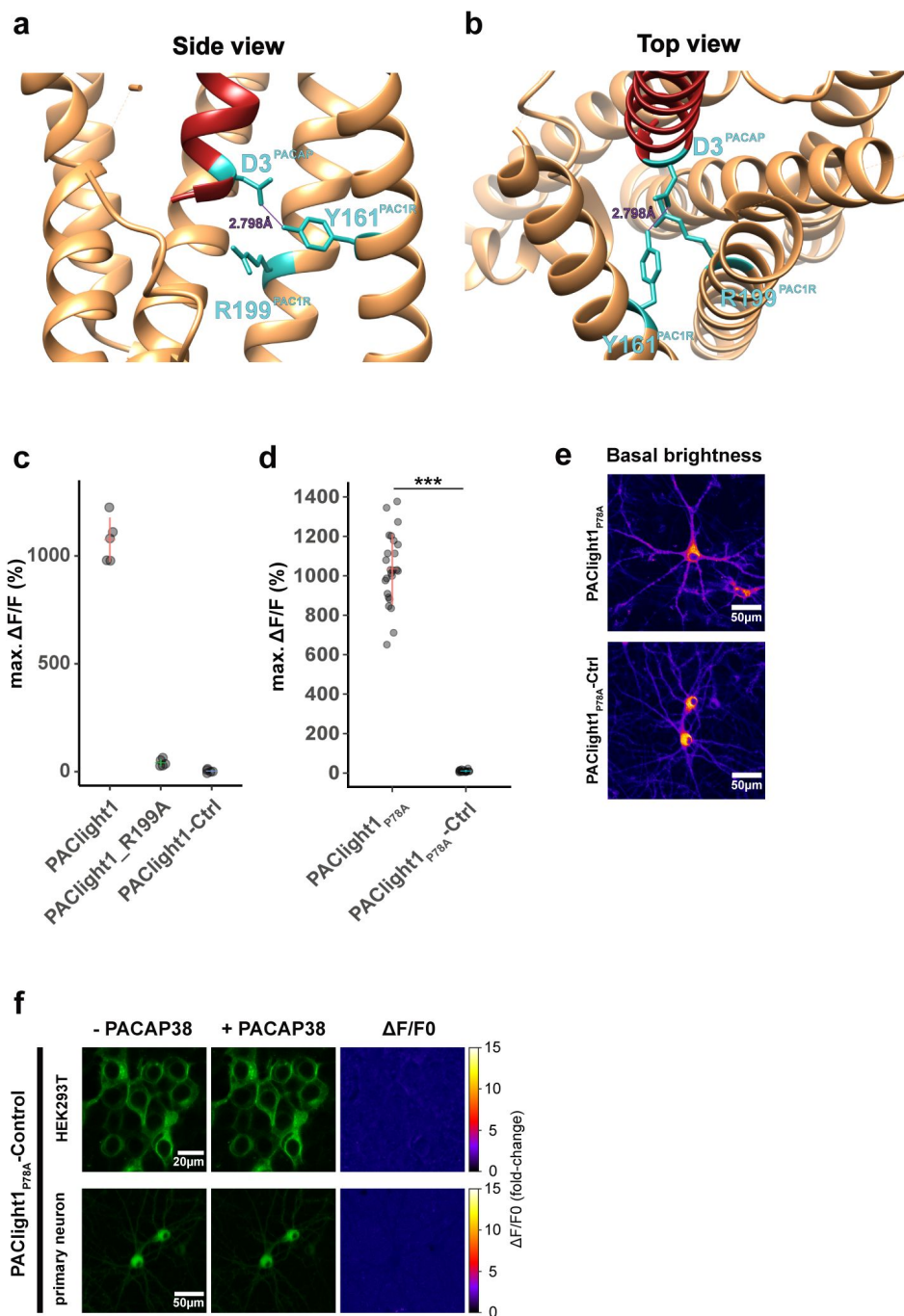
a) Stacked bar plot demonstrating the relative fluorescent response of ECD mutants towards 500nM of VIP followed by 9.09 μ M PACAP₁₋₃₈. The mutant PAC1s contains the naturally occurring splice mutation (deletion) on the ECD. The N60H and N60R were inspired by the analysis of protein structure. The remaining alanine point mutants (right side of bar plot) were inspired by molecular dynamics (see main text). The P78A point mutation on the ECD resulted in excellent improvement of specificity. The highly PACAP-specific mutant was named PAClight1_{P78A}. The data is normalized to the maximum response within each condition. $n=1$ for each mutant. **b)** Visualization of the ECD mutation P78A (cyan) on the backbone of a previously published cryo-EM structure of PAC1Rnull (beige) in interaction with PACAP₁₋₃₈ (dark red) (PDB: 6LPB)²². **c)** Representative confocal micrographs of HEK293T cells expressing the PACAP-specific PAClight1_{P78A} and the VIP-responsive PAClight1 sensors before (left) and after (middle) 10 μ M VIP bath application. Pixel-wise calculation of $\Delta F/F_0$ is shown on the right. Note the complete absence of a fluorescent response to VIP in the PAClight1_{P78A} mutant. **d)** Quantification of the averaged maximum dynamic range of PAClight1_{P78A} ($\Delta F/F_0 = 8.28\%$) vs. PAClight1 ($\Delta F/F_0 = 654\%$) in response to 10 μ M VIP.



Extended Data Figure 3

Characterization of PAClight1_{P78A} coupling to transducer proteins

NanoLuc complementation assays were employed to measure the ability of human SmBiT- PAC1R or SmBiT-PAClight1_{P78A} to recruit LgBiT-miniGs (a), -miniGs (b) or -β-arrestin2 (c) in an agonist-induced manner. Luminescence emission (relative luminescence units, RLU) was monitored during application of agonist (PACAP₁₋₃₈) and was normalized to control cells with application of buffer (vehicle). Dots represent average datapoints, shades represent SEM. n=3 for each condition.



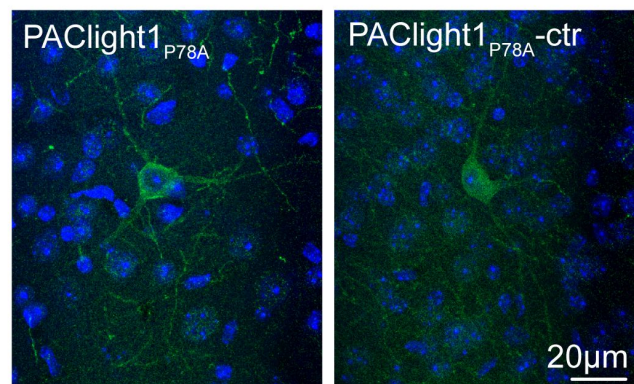
Extended Data Figure 4

Development of a PAClight1_{p78A}-ctrl sensor

a) Side view of the protein structure of the hmPAC1R transmembrane domain (beige) in interaction with PACAP₁₋₃₈ (red). Residues Y161 and R199 of the hmPAC1R, as well as residue D3 of PACAP are highlighted in cyan. The hydrogen bond between D3 and Y161 is indicated with a dashed line. **b)** Top view onto the same protein structures depicted in **a)**. **c)** Maximum dynamic range of HEK293T cells expressing PAClight1 (mean= 1075% $\Delta F/F_0$), PAClight1_R199A (mean= 41.9% $\Delta F/F_0$), and PAClight1-ctrl (mean= 2.87% $\Delta F/F_0$). **d)** Maximum dynamic range of HEK293T cells expressing PAClight1_{p78A} (mean= 1034% $\Delta F/F_0$) and PAClight1_{p78A}-ctrl (mean= 10.44% $\Delta F/F_0$). **e)** Comparison of basal brightness levels of PAClight1_{p78A} and PAClight1_{p78A}-ctrl expressed in rat primary neurons via viral transduction. **f)** Representative examples of PAClight1_{p78A}-ctrl expressed in HEK293T cells (top row) and rat primary neurons (bottom row) before (left column) and after (middle column) bath application of 10 μ M PACAP₁₋₃₈. The right column depicts the pixel-wise calculated $\Delta F/F_0$. No detectable increase in fluorescence is observed.

a

Temporal cortex



Extended Data Figure 5

Validation of PAClight1_{P78A} and PAClight1_{P78A}-ctrl sensors in mammalian brains

Maximum intensity projections of exemplary confocal images of PAClight1_{P78A} and PAClight1_{P78A}-ctrl-expressing neurons in the cortex enhanced with immunostaining and counterstained with DAPI (blue).

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Article and author information

Reto B. Cola

Institute of Pharmacology and Toxicology, University of Zürich, Zürich, Switzerland;

Salome N. Niethammer

Medical University of Vienna, Center for Brain Research, Department for Neuronal Cell Biology, Vienna, Austria

Preethi Rajamannar

Department of Molecular Neuroscience & Department of Molecular Cell Biology Weizmann Institute of Science, Israel

Andrea Gresch

Institute of Pharmacology and Toxicology, University of Zürich, Zürich, Switzerland;

Musadiq A. Bhat

Institute of Pharmacology and Toxicology, University of Zürich, Zürich, Switzerland;

Kevin Assoumou

Department of Cell Physiology and Metabolism, University of Geneva, Geneva, Switzerland;

Elyse T. Williams

Department of Chemistry, University of Zürich, Zürich, Switzerland;

Patrick Hauck

Department of Chemistry, University of Zürich, Zürich, Switzerland;

Nina Hartrampf

Department of Chemistry, University of Zürich, Zürich, Switzerland;

Dietmar Benke

Institute of Pharmacology and Toxicology, University of Zürich, Zürich, Switzerland;;
Neuroscience Center Zurich, University and ETH Zürich, Zürich, Switzerland;

Miriam Stoeber

Department of Cell Physiology and Metabolism, University of Geneva, Geneva, Switzerland;

Gil Levkowitz

Department of Molecular Neuroscience & Department of Molecular Cell Biology Weizmann Institute of Science, Israel
ORCID iD: [0000-0002-3896-1881](https://orcid.org/0000-0002-3896-1881)

Sarah Melzer

Medical University of Vienna, Center for Brain Research, Department for Neuronal Cell Biology, Vienna, Austria

Tommaso Patriarchi

Institute of Pharmacology and Toxicology, University of Zürich, Zürich, Switzerland;;
Neuroscience Center Zurich, University and ETH Zürich, Zürich, Switzerland;

For correspondence: patriarchi@pharma.uzh.ch

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Editors

Reviewing Editor

Amy Andreotti

Iowa State University, Ames, United States of America

Senior Editor

Amy Andreotti

Iowa State University, Ames, United States of America

Reviewer #1 (Public Review):

Summary:

The manuscript "Engineering of PAClight1P78A: A High-Performance Class-B1 GPCR-Based Sensor for PACAP1-38" by Cola et al. presents the development of a novel genetically encoded sensor, PAClight1P78A, based on the human PAC1 receptor. The authors provide a thorough in vitro and in vivo characterization of this sensor, demonstrating its potential utility across various applications in life sciences, including drug development and basic research.

The diverse methods to validate PAClight1P78A demonstrate a comprehensive approach to sensor engineering by combining biochemical characterization with in vivo studies in rodent brains and zebrafish. This establishes the sensor's biophysical properties (e.g., sensitivity, specificity, kinetics, and spectral properties) and demonstrates its functionality in physiologically relevant settings. Importantly, the inclusion of control sensors and the testing of potential intracellular downstream effects such as G-protein activation underscore a careful consideration of specificity and biological impact.

Strengths:

The fundamental development of PAClight1P78A addresses a significant gap in sensors for Class-B1 GPCRs. The iterative design process -starting from PAClight0.1 to the final PAClight1P78A variant - demonstrates compelling optimization. The innovative engineering results in a sensor with a high apparent dynamic range and excellent ligand selectivity, representing a significant advancement in the field. The rigorous in vitro characterization, including dynamic range, ligand specificity, and activation kinetics, provides a critical understanding of the sensor's utility. Including in vivo experiments in mice and zebrafish larvae demonstrates the sensor's applicability in complex biological systems.

Weaknesses:

The manuscript shows that the sensor fundamentally works *in vivo*, albeit in a limited capacity. The titration curves show sensitivity in the nmol range at which endogenous detection might be possible. However, perhaps the sensor is not sensitive enough or there are not any known robust paradigms for PACAP release. A more detailed discussion of the sensors's limitations, particularly regarding *in vivo* applications and the potential for detecting endogenous PACAP release, would be helpful.

There are several experiments with an $n=1$ and other low single-digit numbers. I assume that refers to biological replicates such as mice or culture wells, but it is not well defined. $n=1$ in experimental contexts, particularly in Figure 1, raises significant concerns about the exact dynamic range of the sensor, data reproducibility, and the robustness of conclusions drawn from these experiments. Also, ROI for cell cultures, like in Figure 1, is not well defined. The methods mentioned ROIs were manually selected, which appears very selective, and the values in Figure 1c become unnecessarily questionable. The lack of definition for "ROI" is confusing. Do ROIs refer to cells, specific locations on the cell membrane, or groups of cells? It would be best if the authors could use unbiased methods for image analysis that include the majority of responsive areas or an explanation of why certain ROIs are included or excluded.

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Reviewer #2 (Public Review):

Summary:

The PAClight1 sensor was developed using an approach successful for the development of other fluorescence-based GPCR sensors, which is the complete replacement of the third intracellular loop of the receptor with a circularly-permuted green fluorescent protein. When expressed in HEK cells, this sensor showed good expression and a weak but measurable response to the extracellular presence of PACAP1-38 (a $\Delta F/F_0$ of 43%). Additional mutation near the site of insertion of the linearized GFP, at the C-terminus of the receptor, and within the second intracellular loop produced a final optimized sensor with $\Delta F/F_0$ of >1000%. Finally, screening of mutational libraries that also included alterations in the extracellular ligand-binding domain of the receptor yielded a molecule, PAClight1P78A, that exhibited a high ligand-dependent fluorescence response combined with a high differential sensitivity to PACAP (EC₅₀ 30 nM based on cytometric sorting of stably transfected HEK293 cells) compared to its congener VIP, (with which PACAP shares two highly related receptors, VPAC1 and VPAC2) as well as several unrelated neuropeptides, and significantly slowed activation kinetics by PACAP in the presence of a 10-fold molar excess of the PAC1 antagonist PACAP6-38. A structurally highly similar control construct, PAClight1P78Actl, showed correspondingly similar basal expression in HEK293 cells, but no PACAP-dependent enhancement in fluorescent properties.

PAClight1P78A was expressed in neurons of the mouse cortex via AAV9.hSyn-mediated gene transduction. Slices taken from PAClight1P78A-transfected cortex, but not slices taken from PAClight1P78Actl-transfected cortex exhibited prompt and persistent elevation of $\Delta F/F_0$ after 2 minutes of perfusion with PACAP1-38 which persisted for up to 14 minutes and was statistically significant after perfusion with 3000, but not 300 or 30 nM, of peptide. Likewise, microinfusion of 200 nL of 300 μ M PACAP1-38 into the cortex of optical fiber-implanted freely moving mice elicited a $\Delta F/F_0$ (%) of greater than 15, and significantly higher than that elicited by application of similar concentrations of VIP, CRF, or enkephalin, or vehicle alone. *In vivo* experiments were carried out in zebrafish larvae by the introduction of PAClight1P78A into single-cell stage *Danio rerio* embryos using a Tol2 transposase-based plasmid with a UAS promoter via injection (of plasmid and transposase mRNA), and sorting of post-fertilization embryos using a marker for transgenesis carried in the UAS : PAClight1P78A construct.

Expression of PAClight1P78A was directed to cells in the olfactory bulb which express the fish paralog of the human PAC1 receptor by using the Tg(GnRH3:gal4ff) line, and fluorescent signals were elicited by intracerebroventricular administration of PACAP1-38 at a single concentration (1 mM), which were specific to PACAP and to the presence of PAClight1P78A per se, as controlled by parallel experiments in which PAClight1P78Act1 instead of PAClight1P78A was contained in the transgenic plasmid.

Major strengths and weaknesses of the methods and results:

The report represents a rigorous demonstration of the elicitation of fluorescent signals upon pharmacological exposure to PACAP in nervous system tissue expressing PAClight1P78A in both mammals (mice) and fish (zebrafish larvae). Figure 4d shows a change in GFP fluorescence activation by PACAP occurring several seconds after the cessation of PACAP perfusion over a two-minute period, and its persistence for several minutes following. One wonders if one is apprehending the graphical presentation of the data incorrectly, or if the activation of fluorescence efficiency by ligand presentation is irreversible in this context, in which case the utility of the probe as a real-time indicator, in vivo, of released peptide might be diminished.

Appraisal of achievement of aims, and data support of conclusions:

Small cavils with controls are omitted for clarity; the larger issue of appraisal of results based on the scope of the designed experiments is discussed in the section below. An interesting question related to the time dependence of the PACAP-elicited activation of PAClight1P87A is its onset and reversibility, and additional data related to this would be welcome.

Discussion of the impact of the work, and utility of the methods and data:

Increasingly, neurotransmitter function may be observed in vivo, rather than by inferring in vivo function from in vitro, in cellular, or ex vivo experimentation. This very valuable report discloses the invention of a genetically encoded sensor for the class B1 GPCR PAC1. PAC1 is the major receptor for the neuropeptide PACAP, which in turn is a major neurotransmitter involved in brain response to psychogenic stress, or threat, in vertebrates as diverse as mammals and fishes. If this sensor possesses the sensitivity to detect endogenously released PACAP in vivo it will indeed be an impactful tool for understanding PACAP neurotransmission (and indeed PACAP action in general, in immune and endocrine compartments as well) in future experiments.

However, the sensor has not yet been used to detect endogenously released PACAP. Until this has been done, one cannot answer the question as to whether the levels of exogenously perfused/administered PACAP used here merely to calibrate the sensor's sensitivity are indeed unphysiologically high. If endogenous PACAP levels don't get that high, then the sensor will not be useful for its intended purpose. The authors should address this issue and allude to what kind of experiments would need to be done in order to detect endogenous PACAP release in living tissue in intact animals. The authors could comment upon the success of other GPCR sensors that have been used to observe endogenous ligand release, and where along the pathway to becoming a truly useful reagent this particular sensor is.

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Reviewer #3 (Public Review):

Summary:

The manuscript introduces PAClight1P78A, a novel genetically encoded sensor designed to facilitate the study of class-B1 G protein-coupled receptors (GPCRs), focusing on the human

PAC1 receptor. Addressing the significant challenge of investigating these clinically relevant drug targets, the sensor demonstrates a high dynamic range, excellent ligand selectivity, and rapid activation kinetics. It is validated across a variety of experimental contexts including in vitro, ex vivo, and in vivo models in mice and zebrafish, showcasing its utility for high-throughput screening, basic research, and drug development efforts related to GPCR dynamics and pharmacology.

Strengths:

The innovative design of PAClight1P78A successfully bridges a crucial gap in GPCR research by enabling real-time monitoring of receptor activation with high specificity and sensitivity. The extensive validation across multiple models emphasizes the sensor's reliability and versatility, promising significant contributions to both the scientific understanding of GPCR mechanisms and the development of novel therapeutics. Furthermore, by providing the research community with detailed methodologies and access to the necessary viral vectors and plasmids, the authors ensure the sensor's broad applicability and ease of adoption for a wide range of studies focused on GPCR biology and drug targeting.

Weaknesses

To further strengthen the manuscript and validate the efficacy of PAClight1P78A as a selective PACAP sensor, it is crucial to demonstrate the sensor's ability to detect endogenous PACAP release in vivo under physiological conditions. While the current data from artificial PACAP application in mouse brain slices and microinfusion in behaving mice provide foundational insights into the sensor's functionality, these approaches predominantly simulate conditions with potentially higher concentrations of PACAP than naturally occurring levels.

Although the sensor's specificity for the PAC1 receptor and its primary ligand is a pivotal achievement, exploring its potential application to other GPCRs within the class-B1 family or broader categories could enhance the manuscript's impact, suggesting ways to adapt this technology for a wider array of receptor studies. Additionally, while the sensor's performance is convincingly demonstrated in short-term experiments, insights into its long-term stability and reusability in more prolonged or repeated measures scenarios would be valuable for researchers interested in chronic studies or longitudinal behavioral analyses. Addressing these aspects could broaden the understanding of the sensor's practical utility over extended research timelines.

Furthermore, the current in vivo experiments involving microinfusion of PACAP near sensor-expressing areas in behaving mice are based on a relatively small sample size ($n=2$), which might limit the generalizability of the findings. Increasing the number of subjects in these experimental groups would enhance the statistical power of the results and provide a more robust assessment of the sensor's in vivo functionality. Expanding the sample size will not only validate the findings but also address potential variability within the population, thereby reinforcing the conclusions drawn from these crucial experiments.

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